



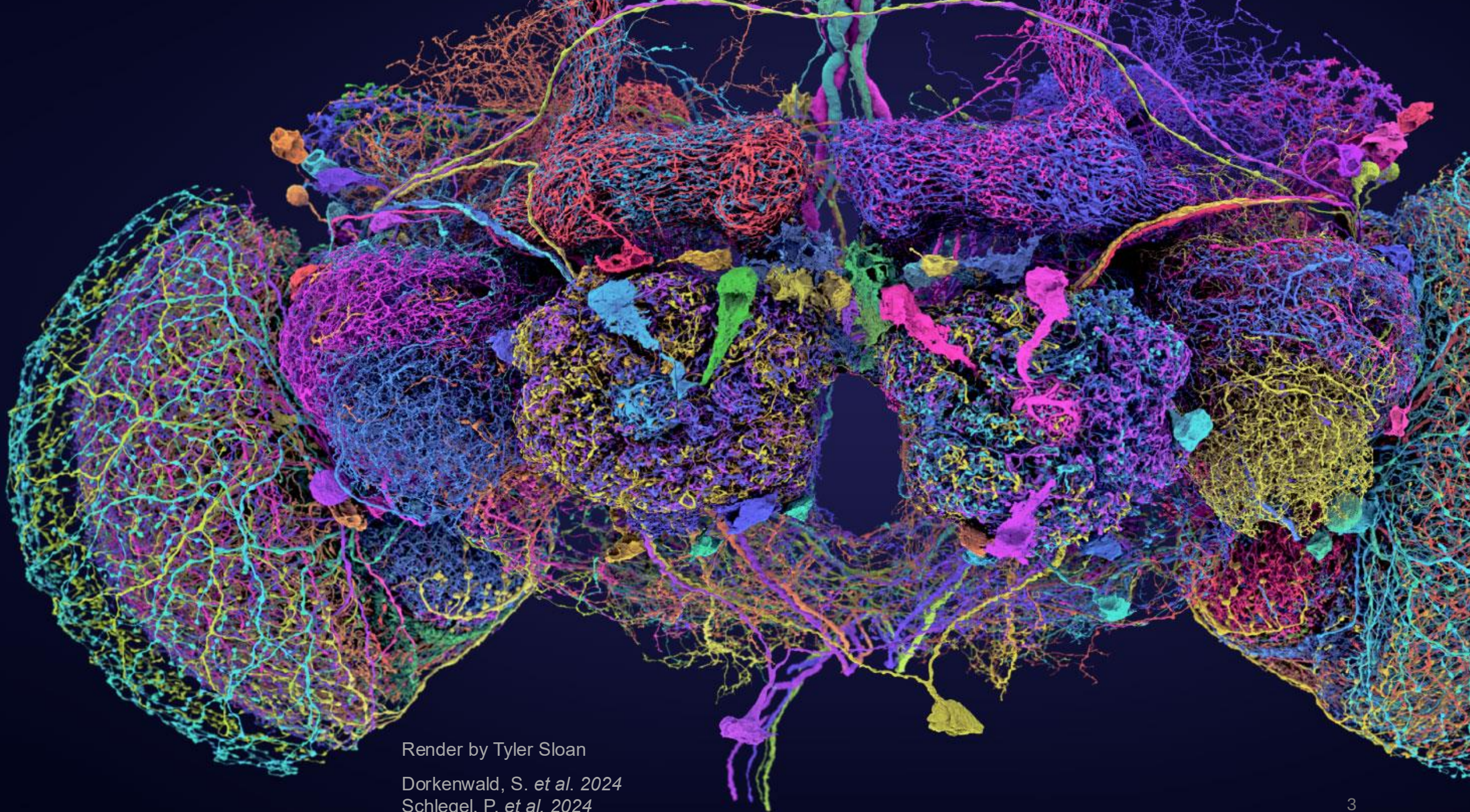
From connectomes to computation: predicting neural function with machine learning

Janne Lappalainen
University of Tübingen, Tübingen AI Center
& HHMI Janelia Research Campus

July 17th, 2025
Cajal NeuroAI Course

Intro

Core Results
Summary
Project ideas



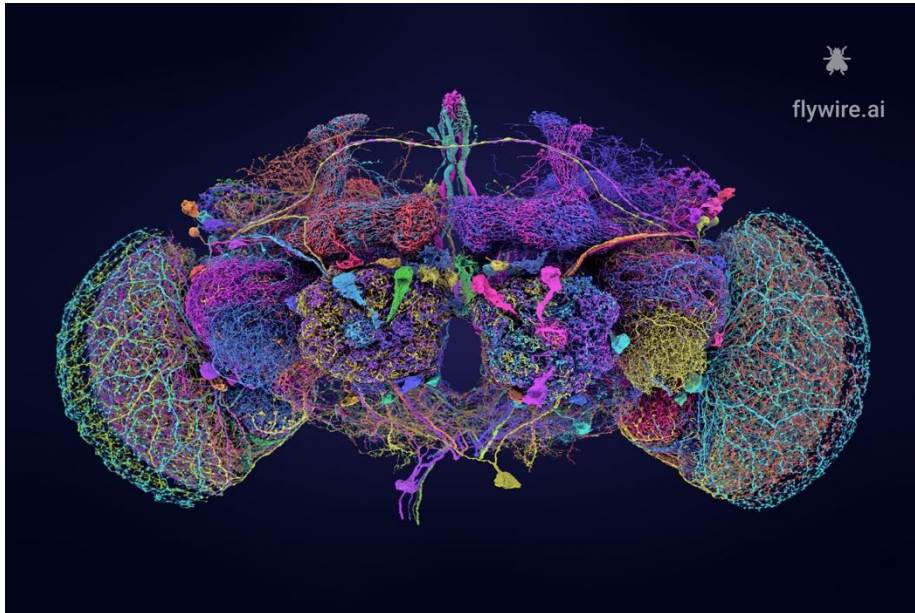
Render by Tyler Sloan

Dorkenwald, S. *et al.* 2024

Schlegel, P. *et al.* 2024

Understand computation in detail and brain-wide

Fruit fly brain structure



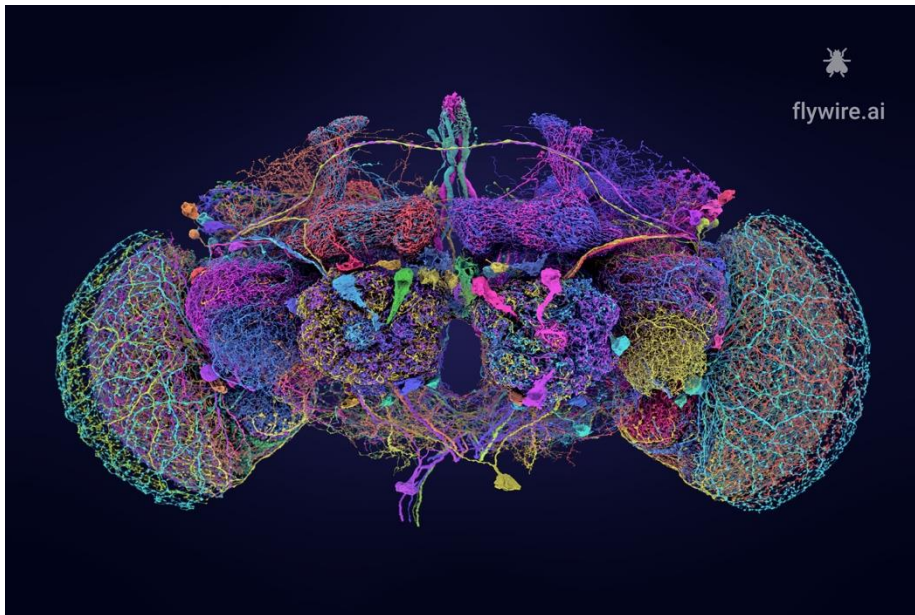
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Schlegel, P. *et al.* (2024)

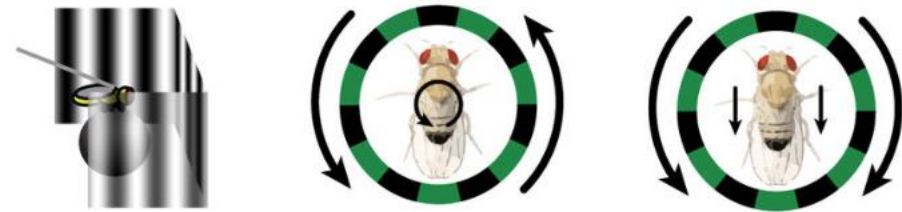
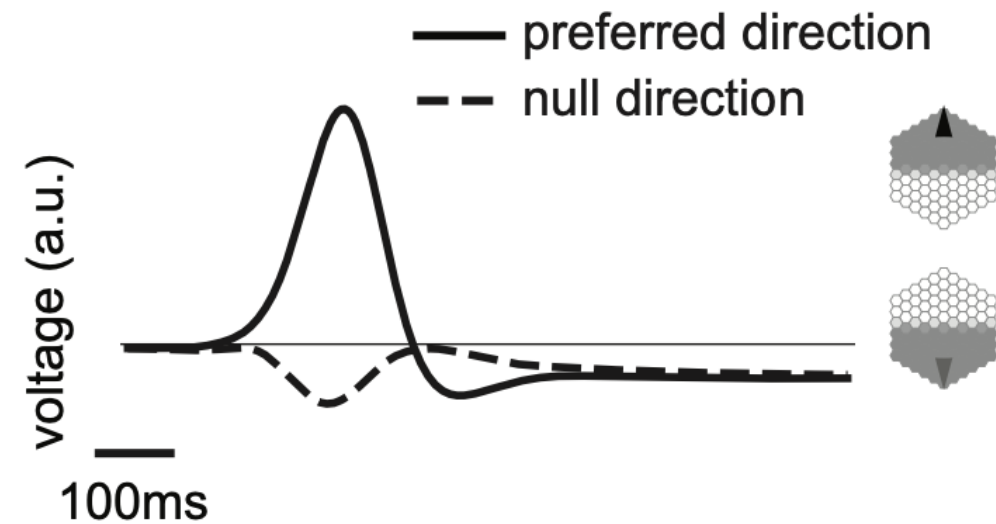
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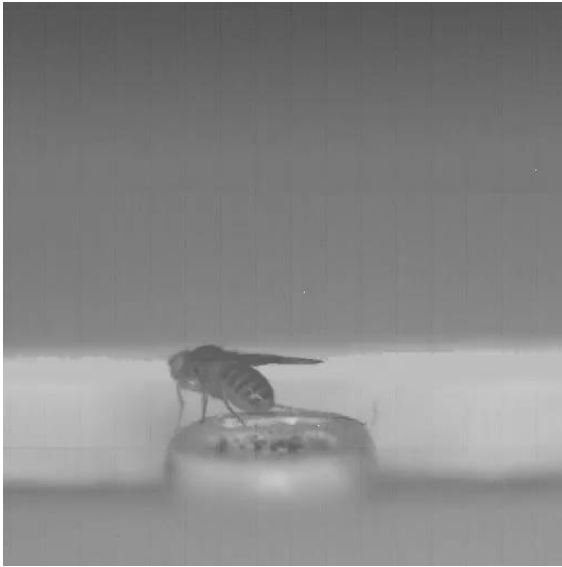


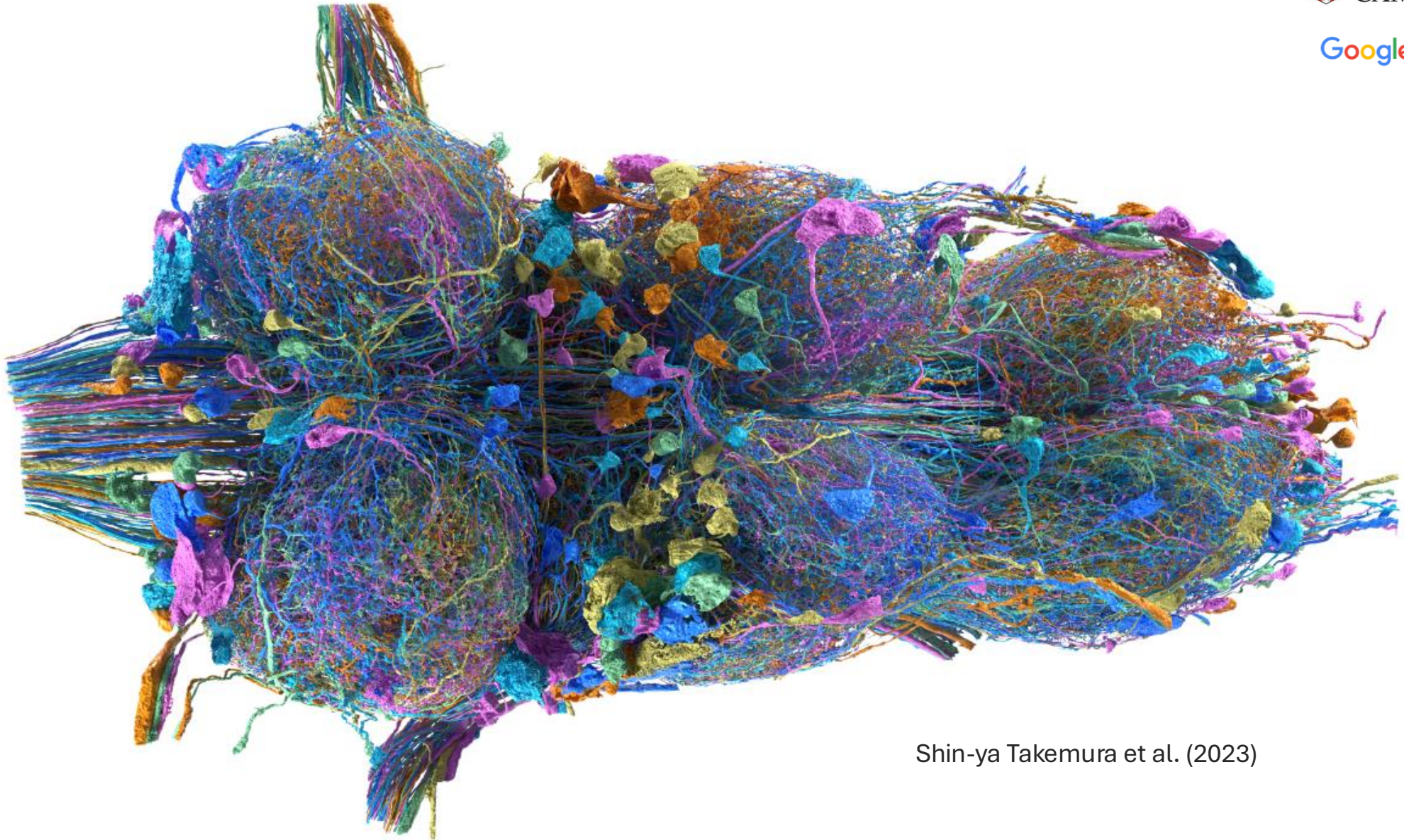
Render by Tyler Sloan
Dorkenwald, S. *et al.* (2024)
Schlegel, P. *et al.* (2024)

Motion computing neuron response

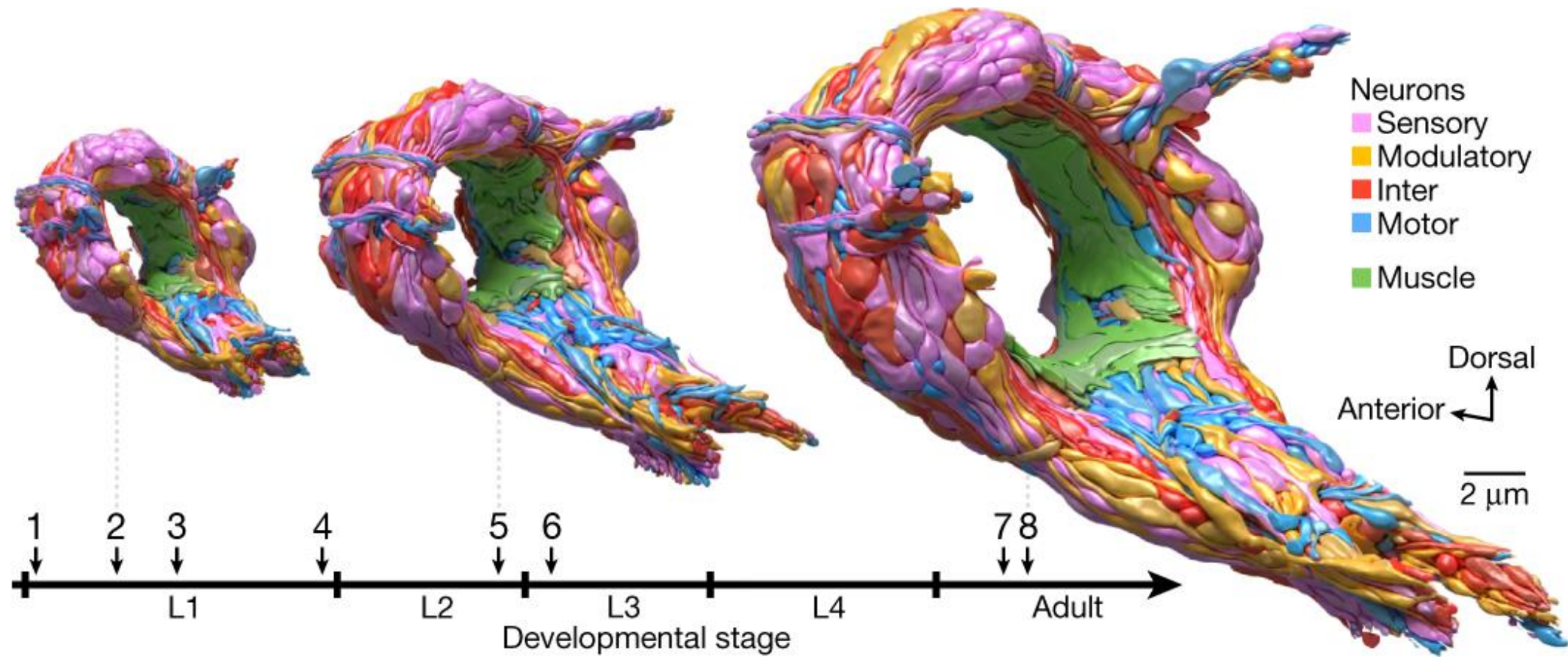


Fly brain is efficient: 140.000 neurons to survive





Shin-ya Takemura et al. (2023)



Witvliet, D. *et al.* (2021)

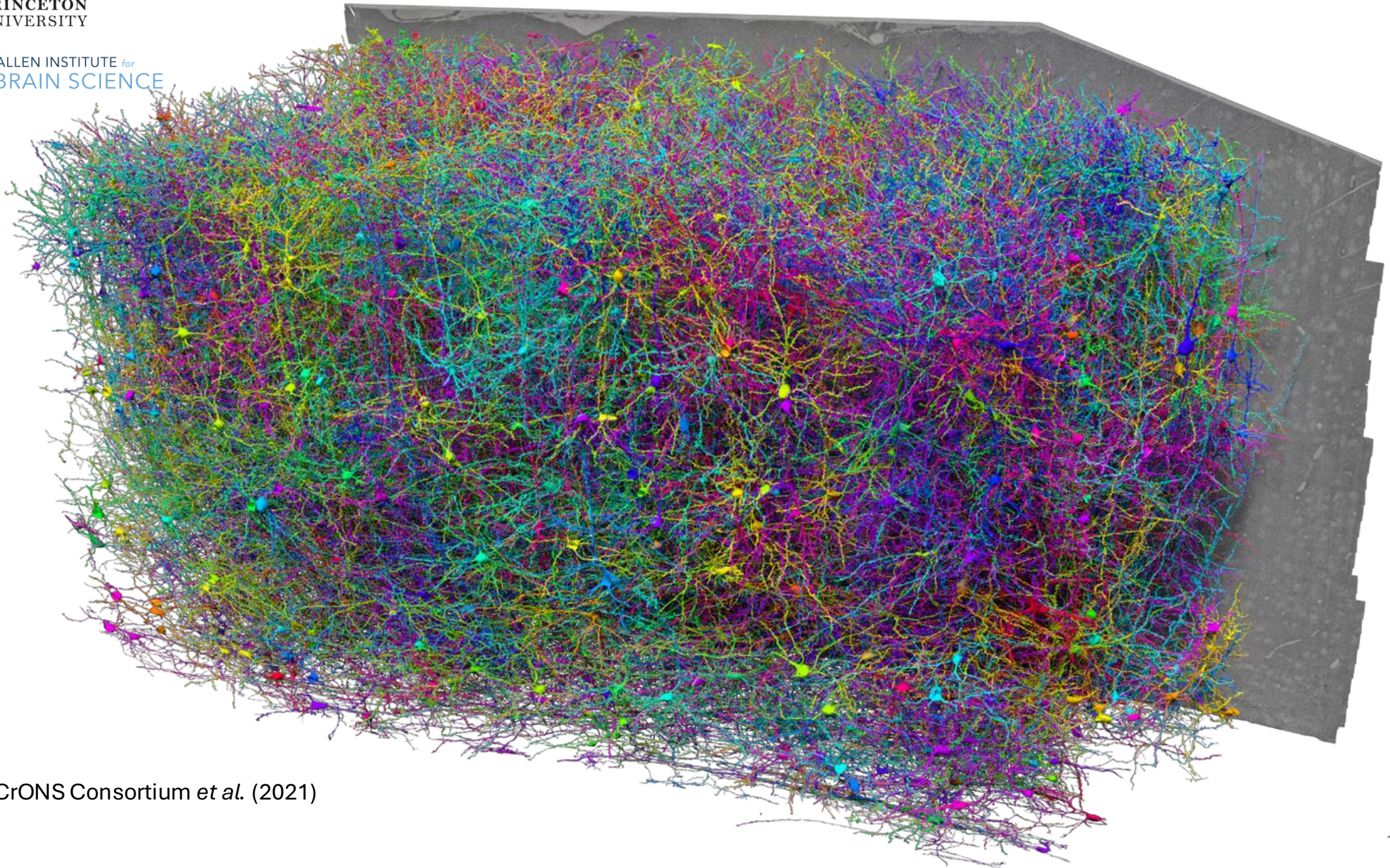


Vishwanathan, A. et al. (2024)

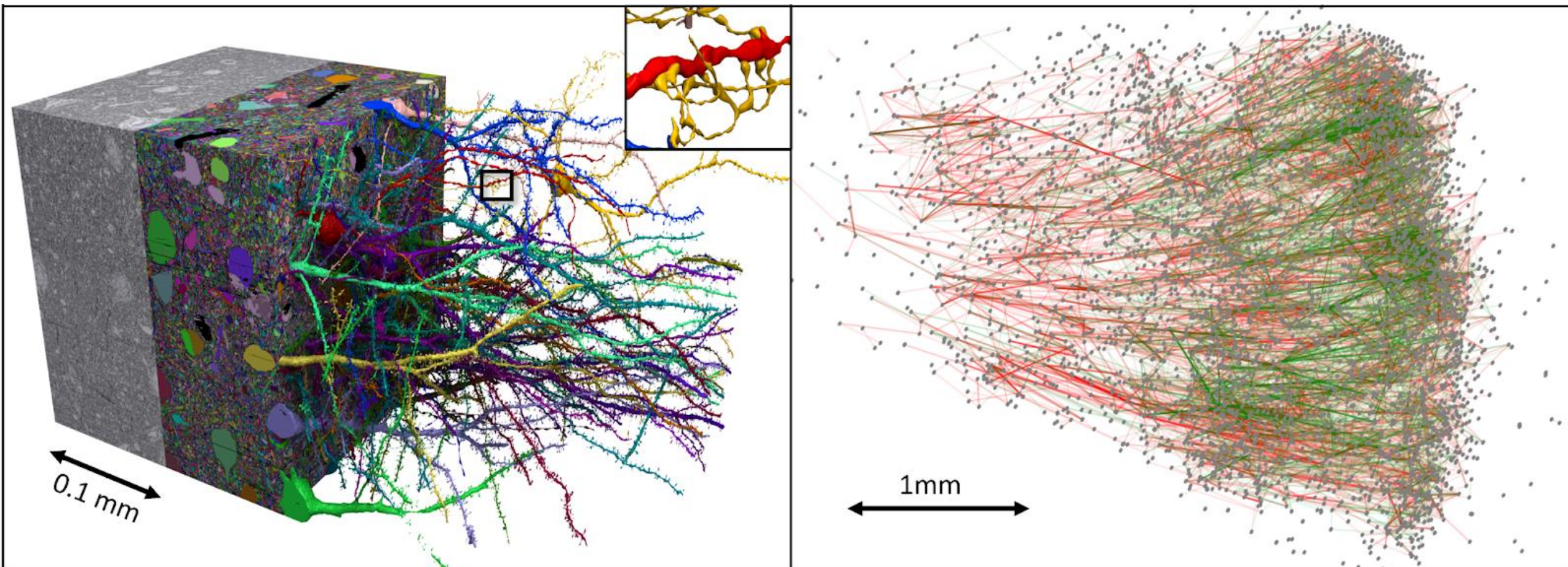
PRINCETON
UNIVERSITY

UC DAVIS
UNIVERSITY OF CALIFORNIA

Weill Cornell
Medicine

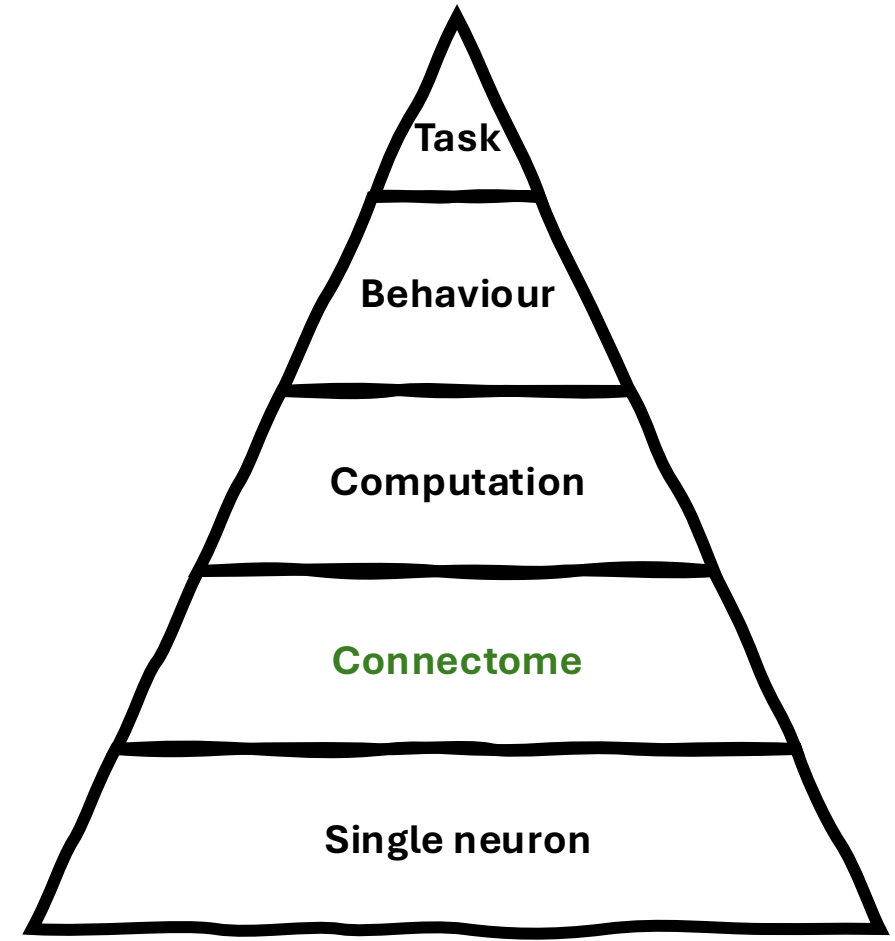


MICrONS Consortium *et al.* (2021)



Shapson-Coe, A. *et al.* (2021)

Challenge: From connectomes to computation

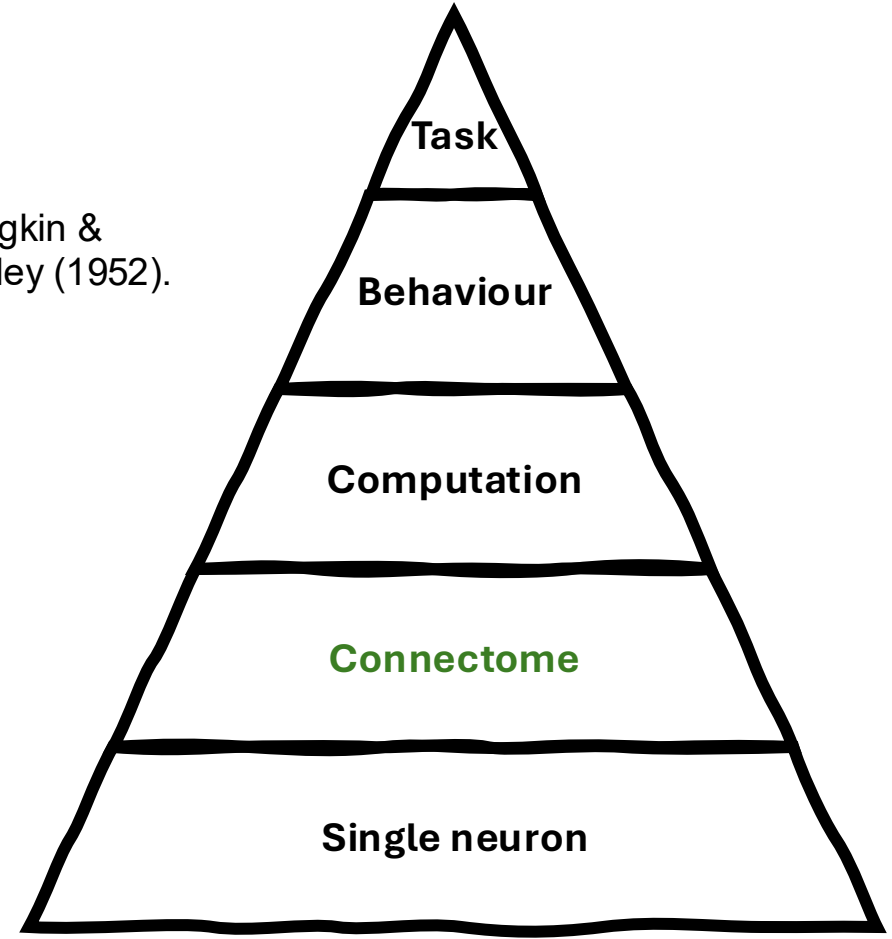


Challenge: From connectomes to computation

We can model single neurons for many years

$$C_m \frac{dV}{dt} = I - g_{\text{Na}} m^3 h (V - V_{\text{Na}}) - g_{\text{K}} n^4 (V - V_{\text{K}}) - g_L (V - V_L)$$

Hodgkin &
Huxley (1952).



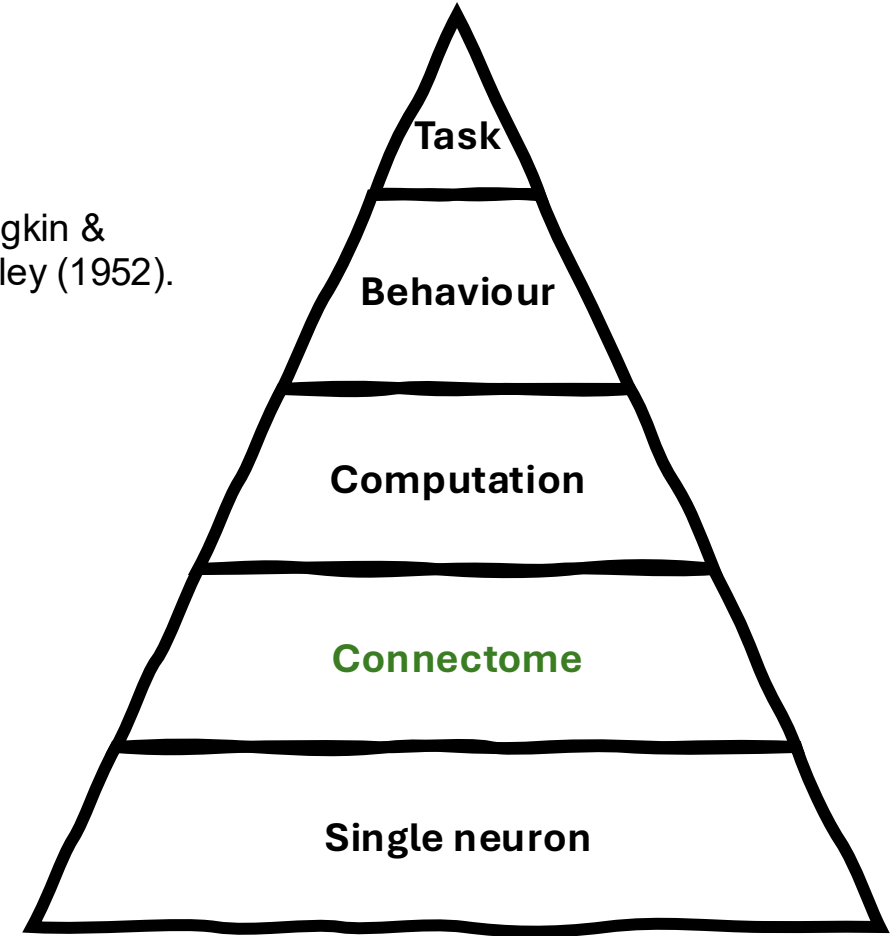
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Huxley (1952).

- Parameters unknown



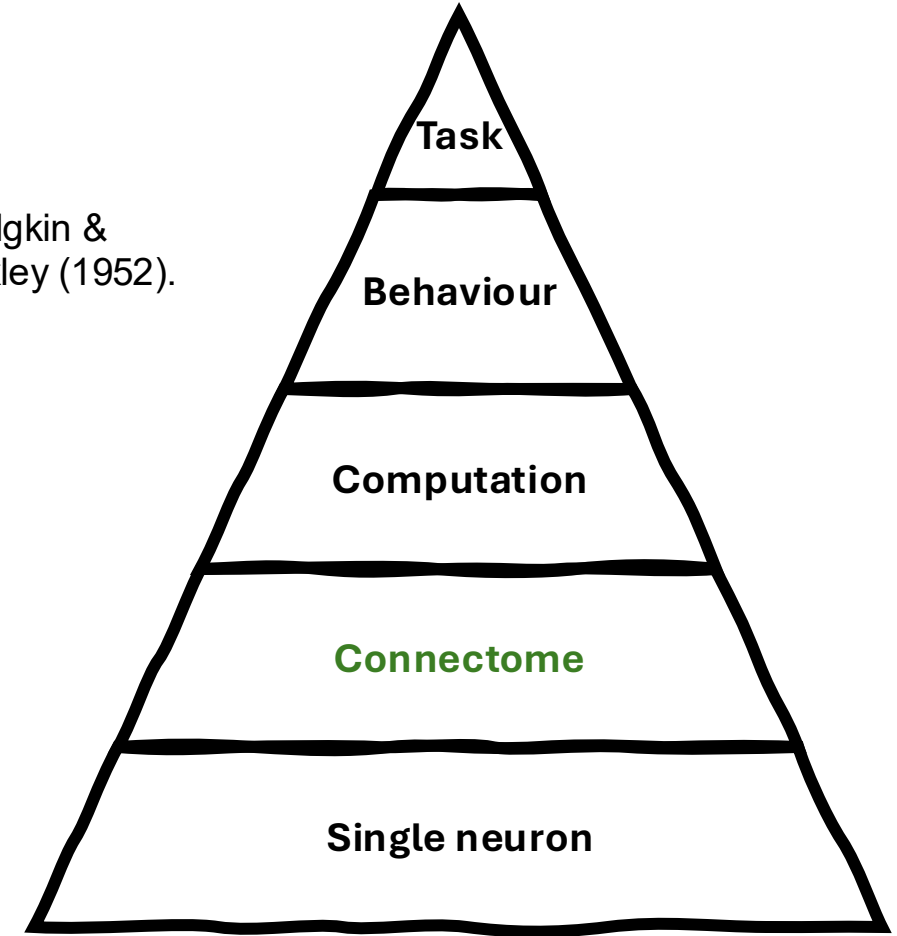
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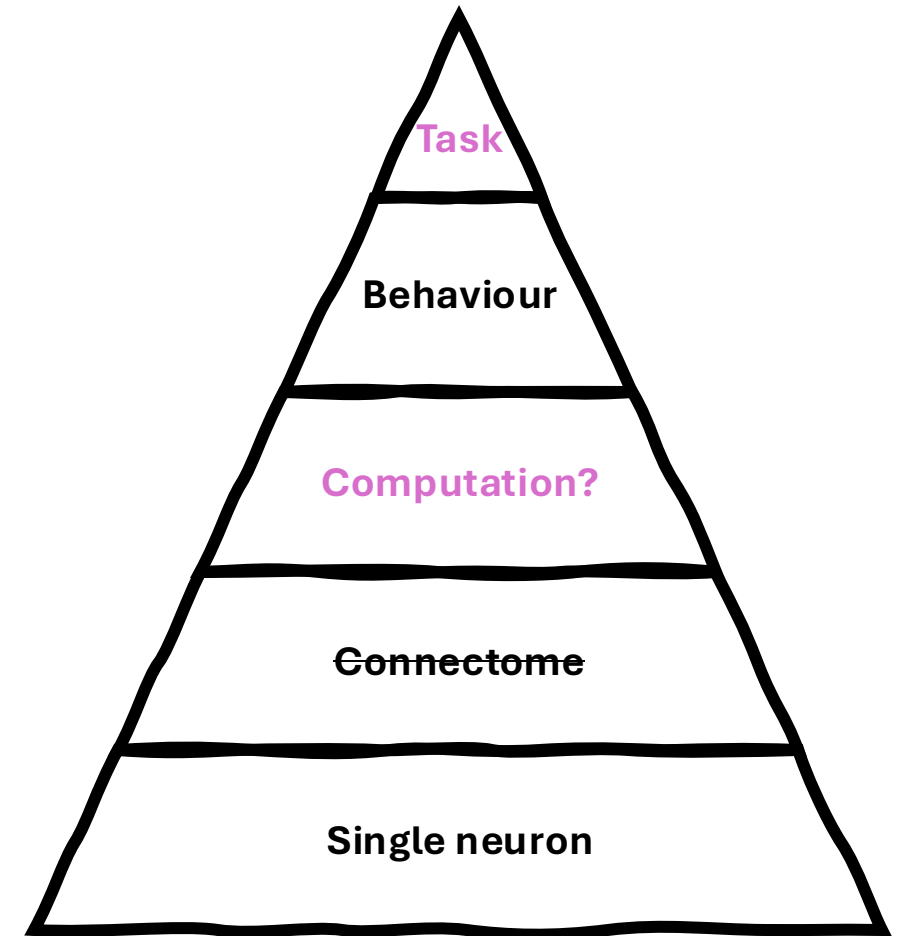
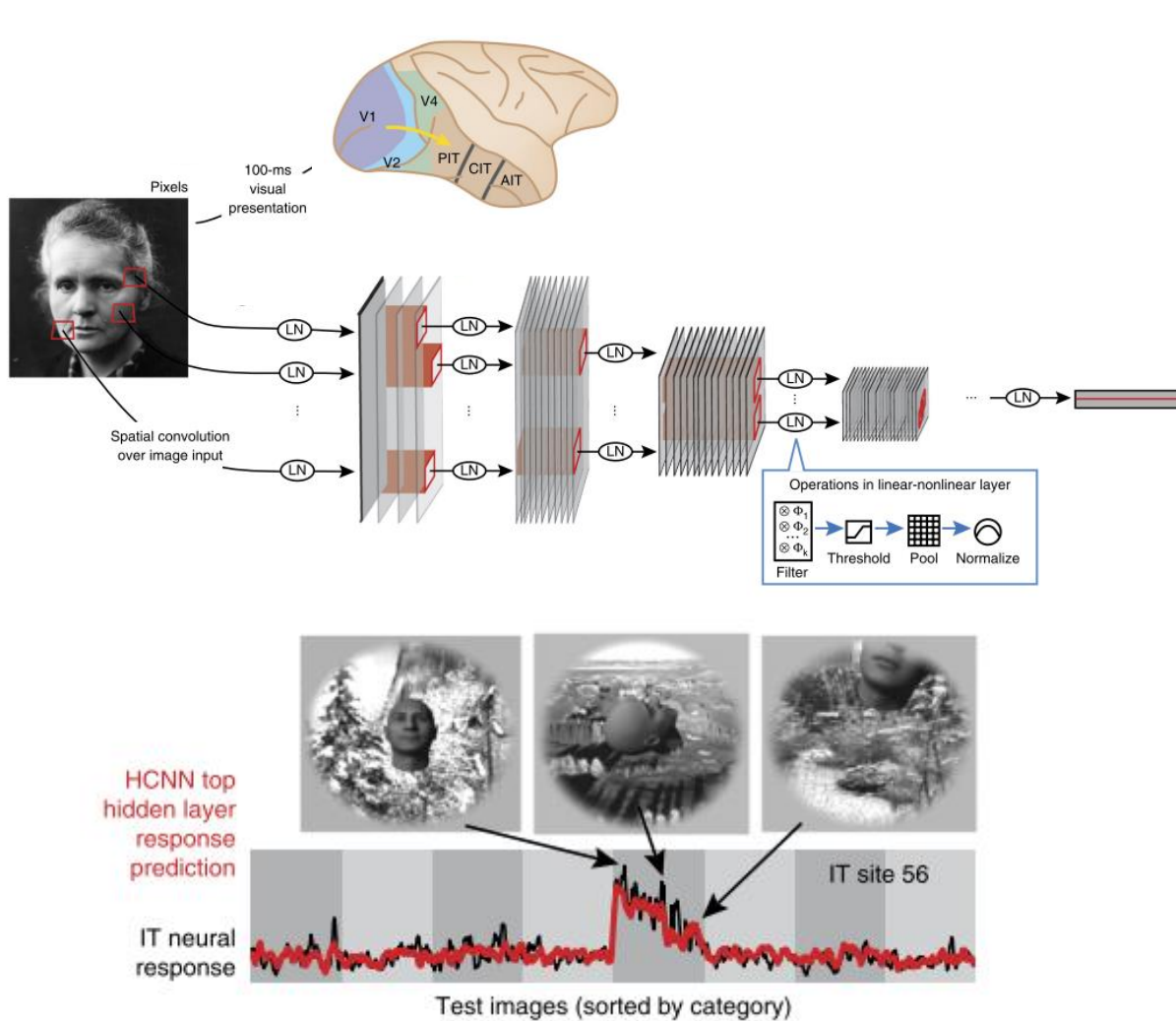
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Hodgkin &
Huxley (1952).

- Parameters unknown
- Inference



Challenge: From connectomes to computation



adapted from Yamins and DiCarlo (2016)

Challenge: From connectomes to computation

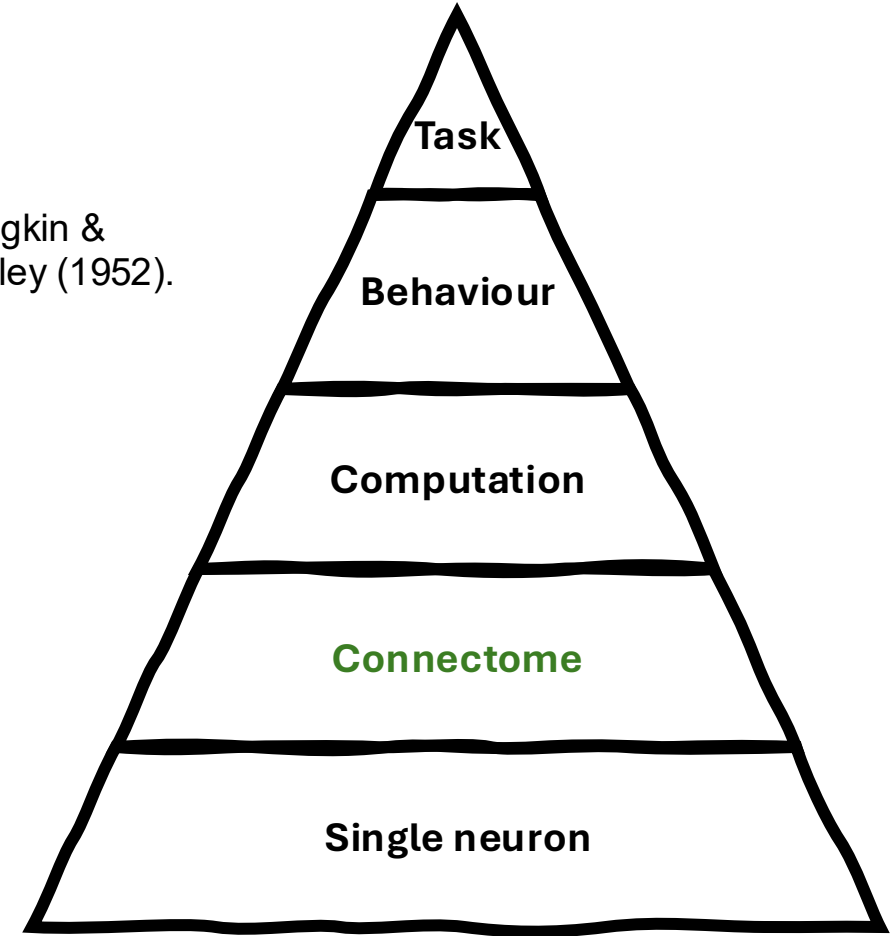
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$$C_m \frac{dV}{dt} = I - g_{Na} m^3 h (V - V_{Na}) - g_K n^4 (V - V_K) - g_L (V - V_L)$$

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- Parameters unknown
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The same network architecture with different parameters can:



Challenge: From connectomes to computation

We can model single neurons for many years

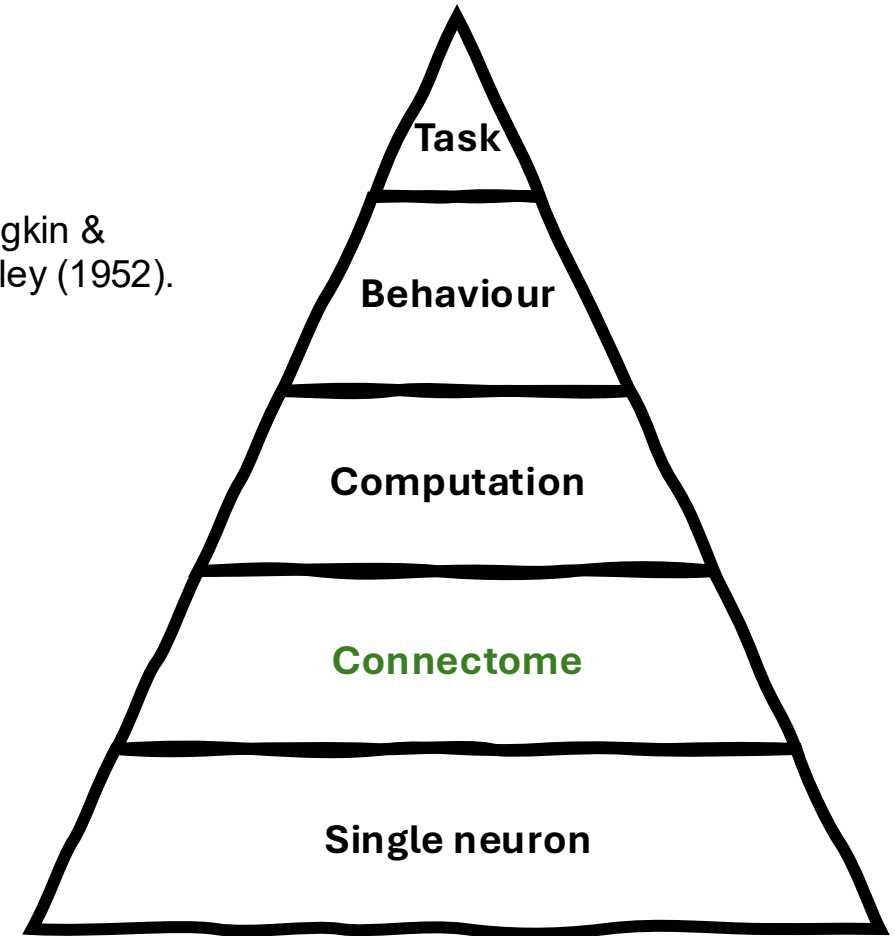
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- Parameters unknown
- Inference

The same network architecture with different parameters can:

- Produce **similar** functions (Prinz et al. 2004)



Challenge: From connectomes to computation

We can model single neurons for many years

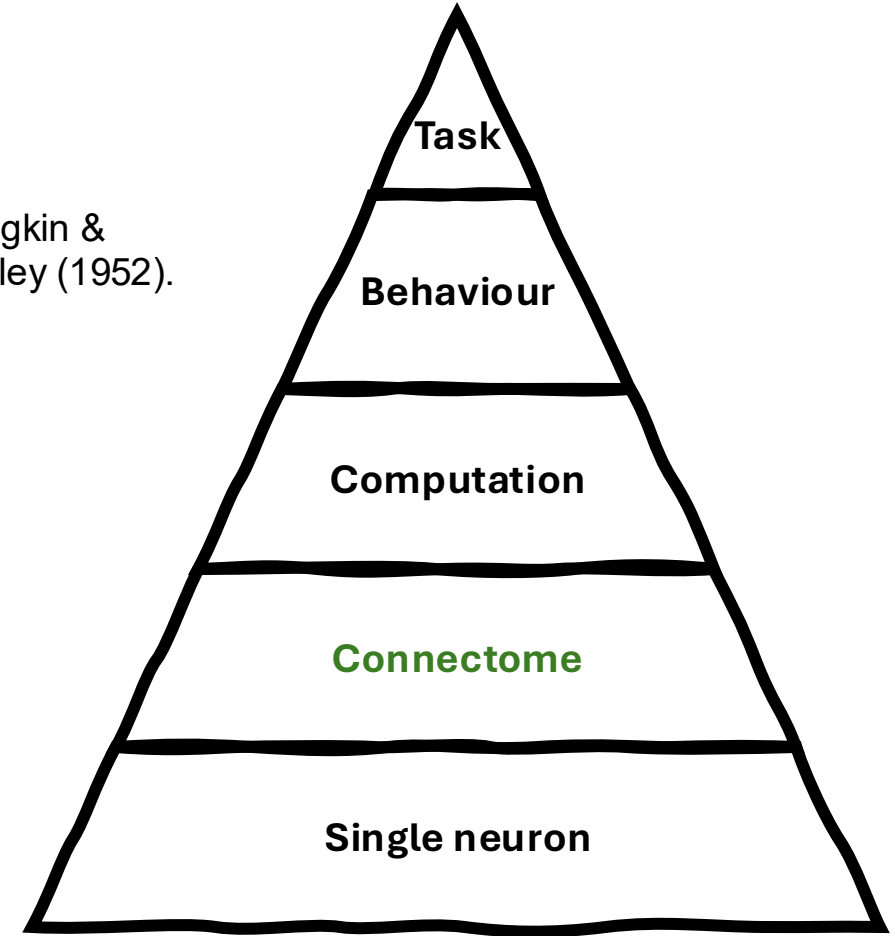
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Hodgkin &
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- Parameters unknown
- Inference

The same network architecture with different parameters can:

- Produce **similar** functions (Prinz et al. 2004)
- Approximate **different** functions (Hornik 1989)



Challenge: From connectomes to computation

We can model single neurons for many years

$$C_m \frac{dV}{dt} = I - g_{Na} m^3 h (V - V_{Na}) - g_K n^4 (V - V_K) - g_L (V - V_L)$$

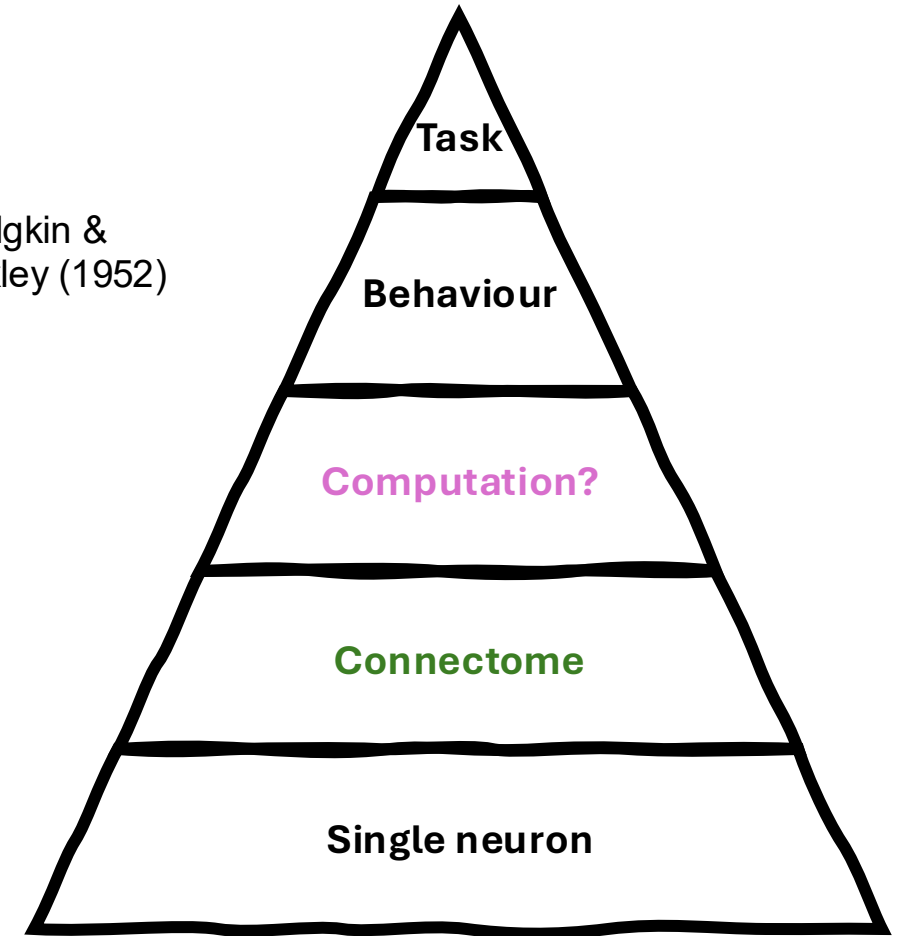
Hodgkin & Huxley (1952)

- Parameters unknown
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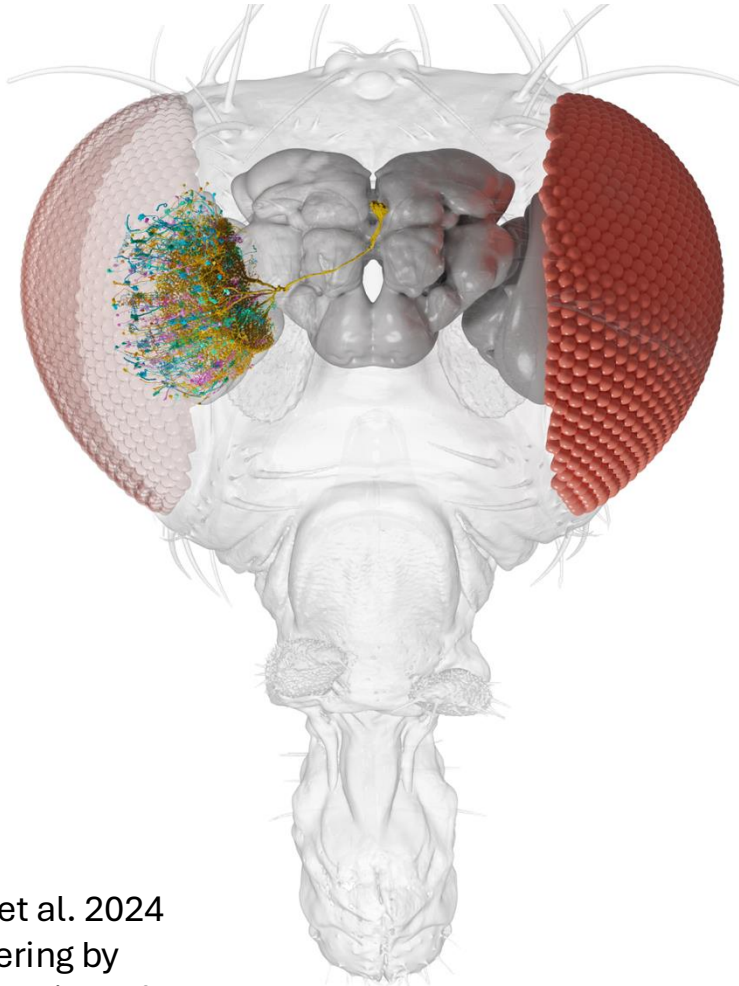
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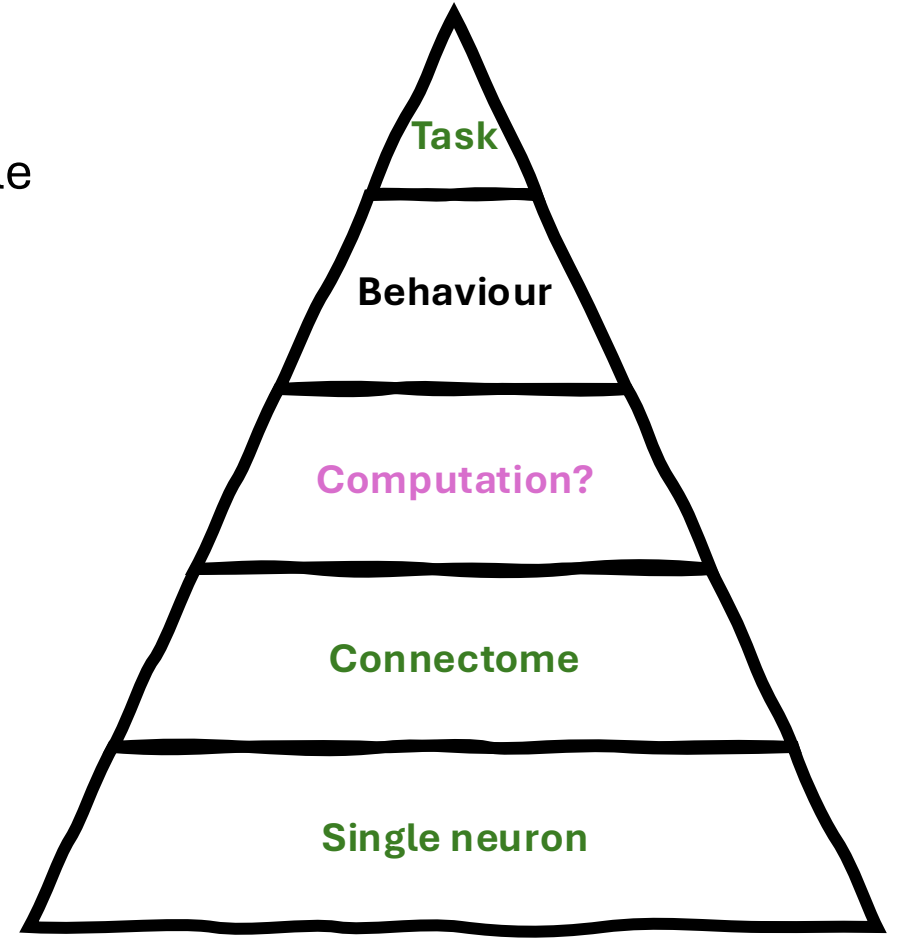
Can we find parameters that constrain computation with connectomes and task meaningfully?



The fly visual system opportunity

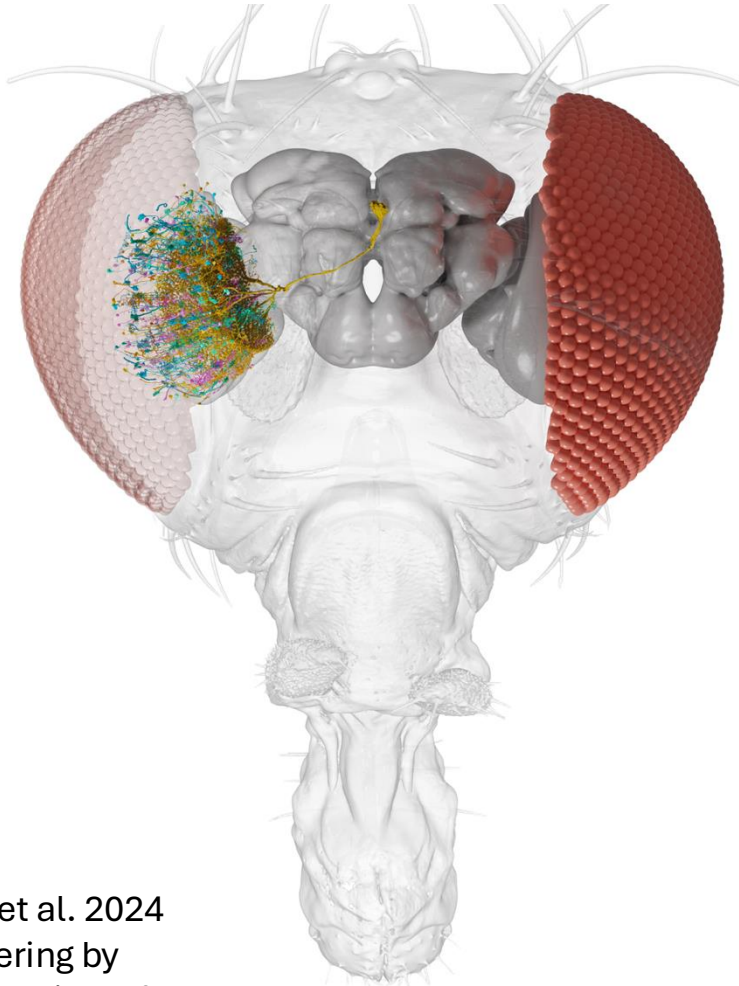


- Connectome available

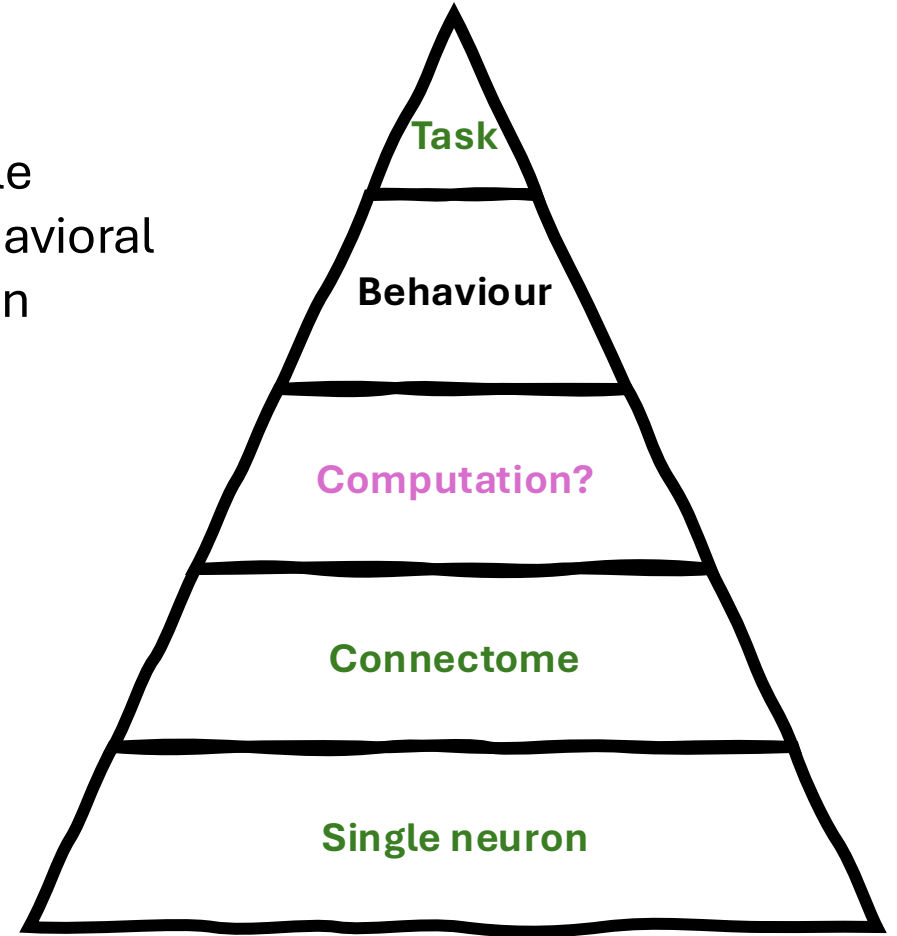


Nern et al. 2024
Rendering by
Siwanowicz I. & Loesche F

The fly visual system opportunity

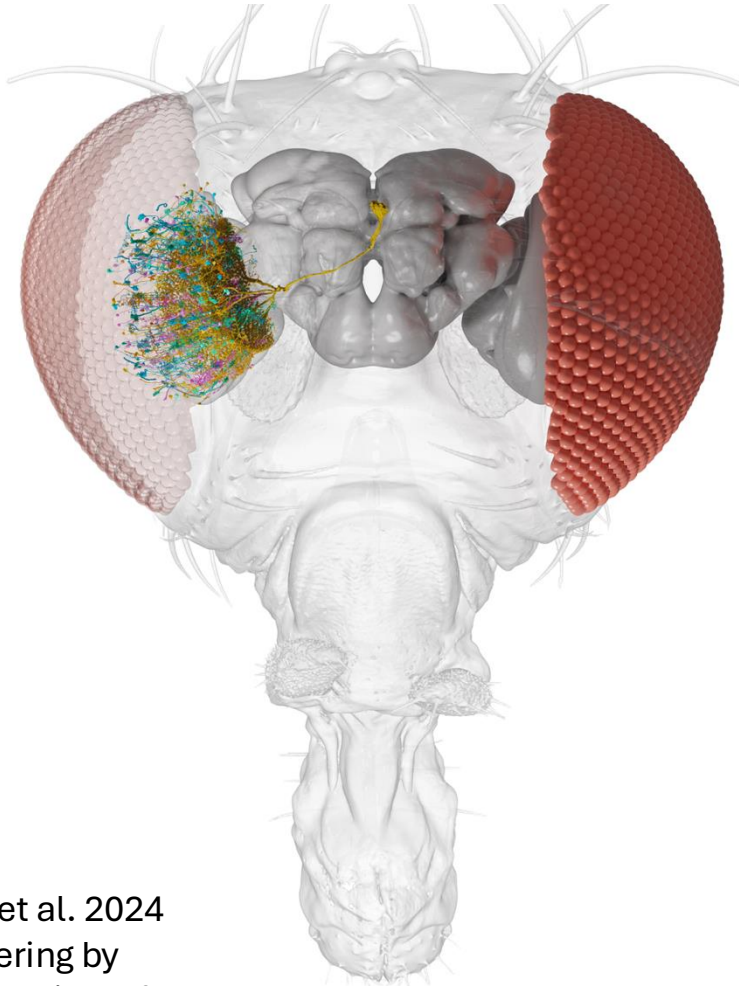


- Connectome available
- Circuit with clear behavioral task: motion detection

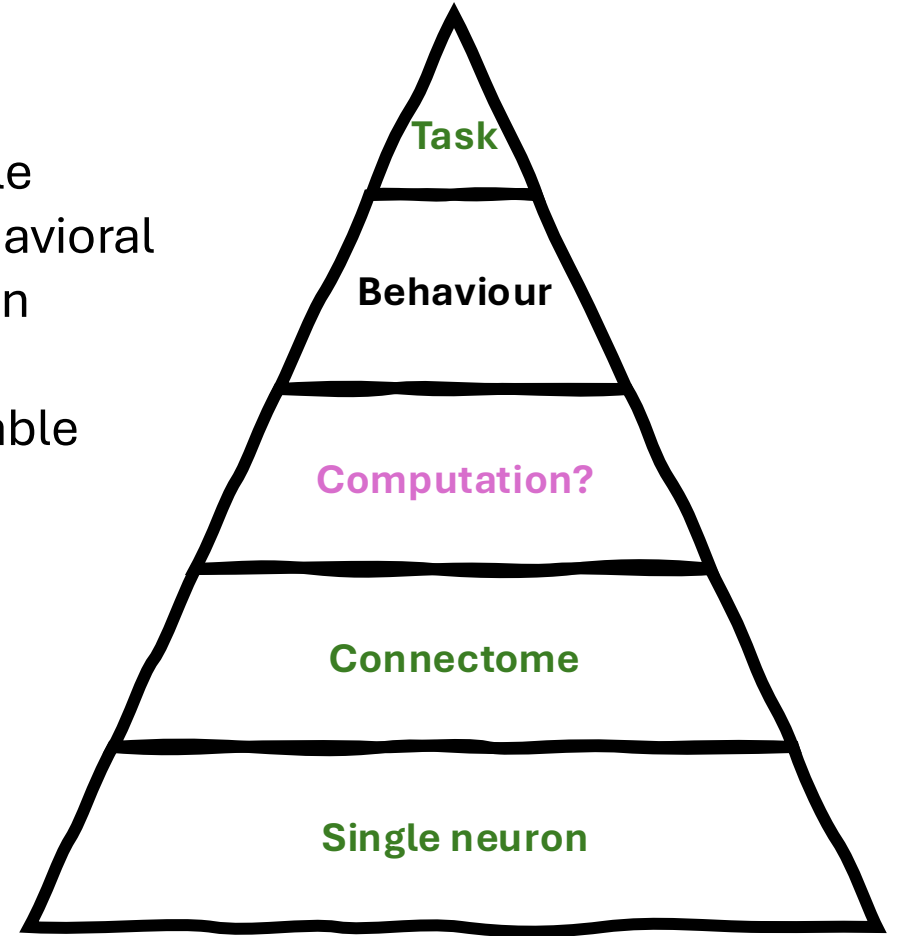


Nern et al. 2024
Rendering by
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The fly visual system opportunity

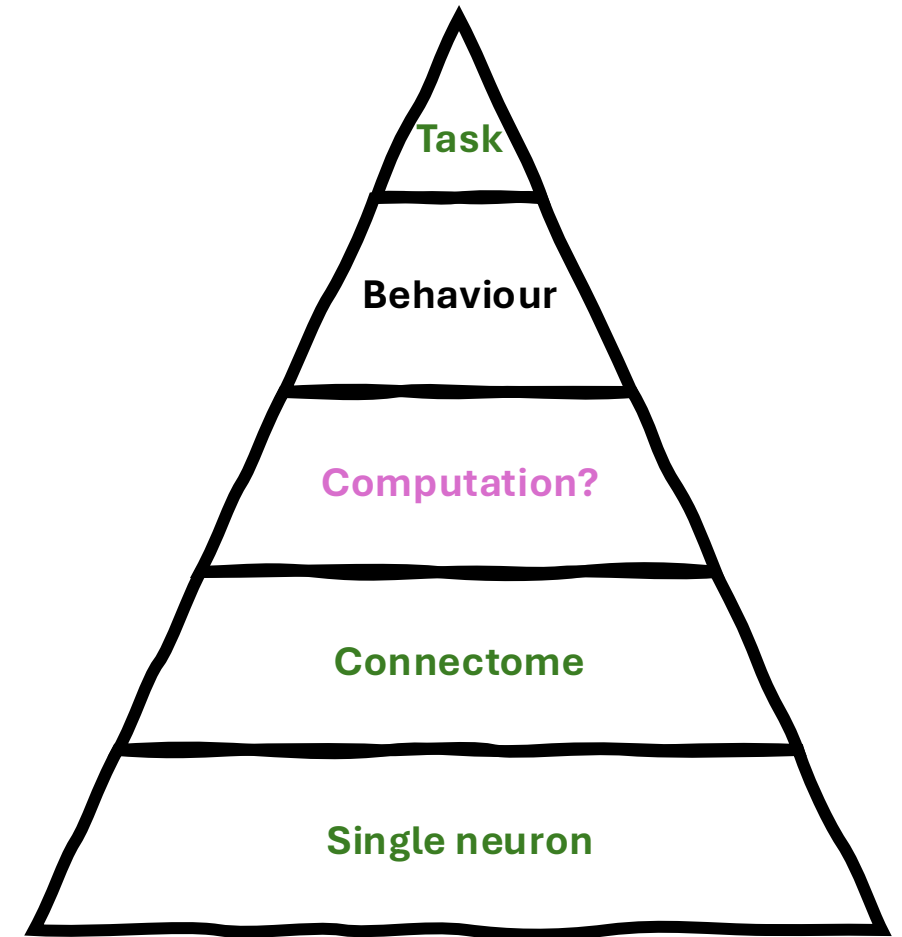
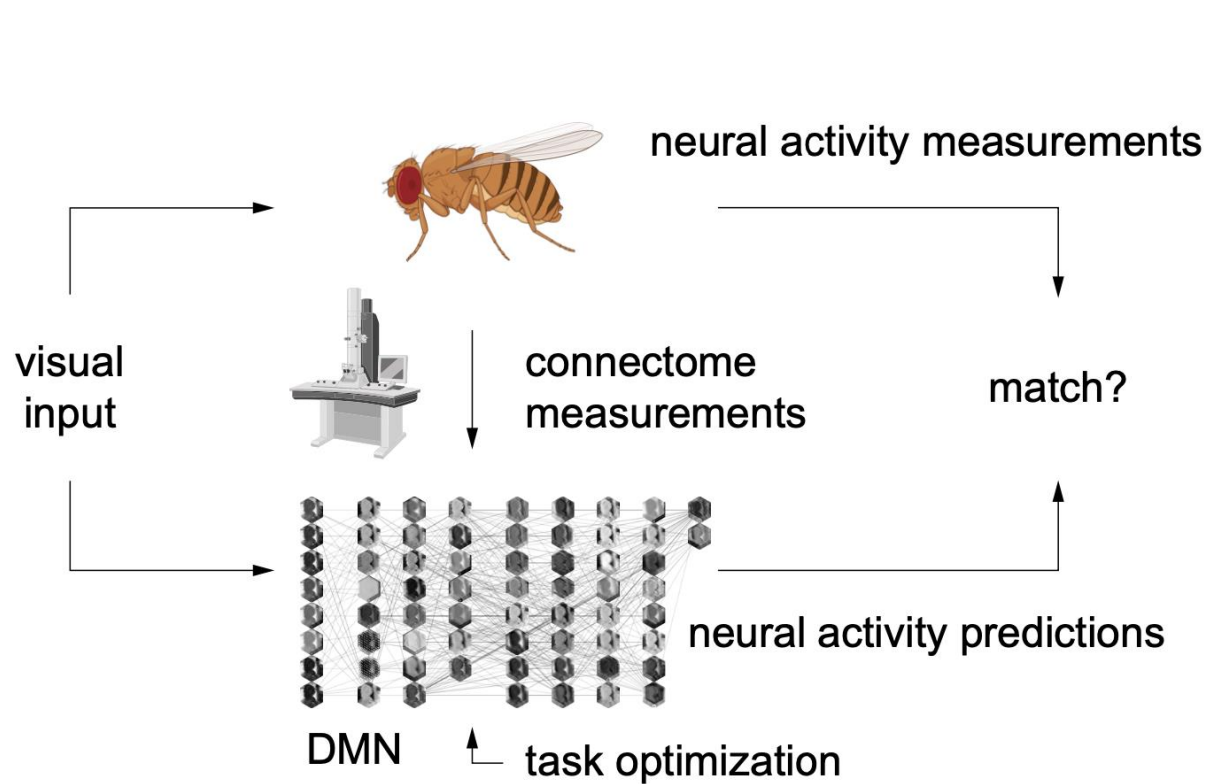


- Connectome available
- Circuit with clear behavioral task: motion detection
- Neural activity measurements available



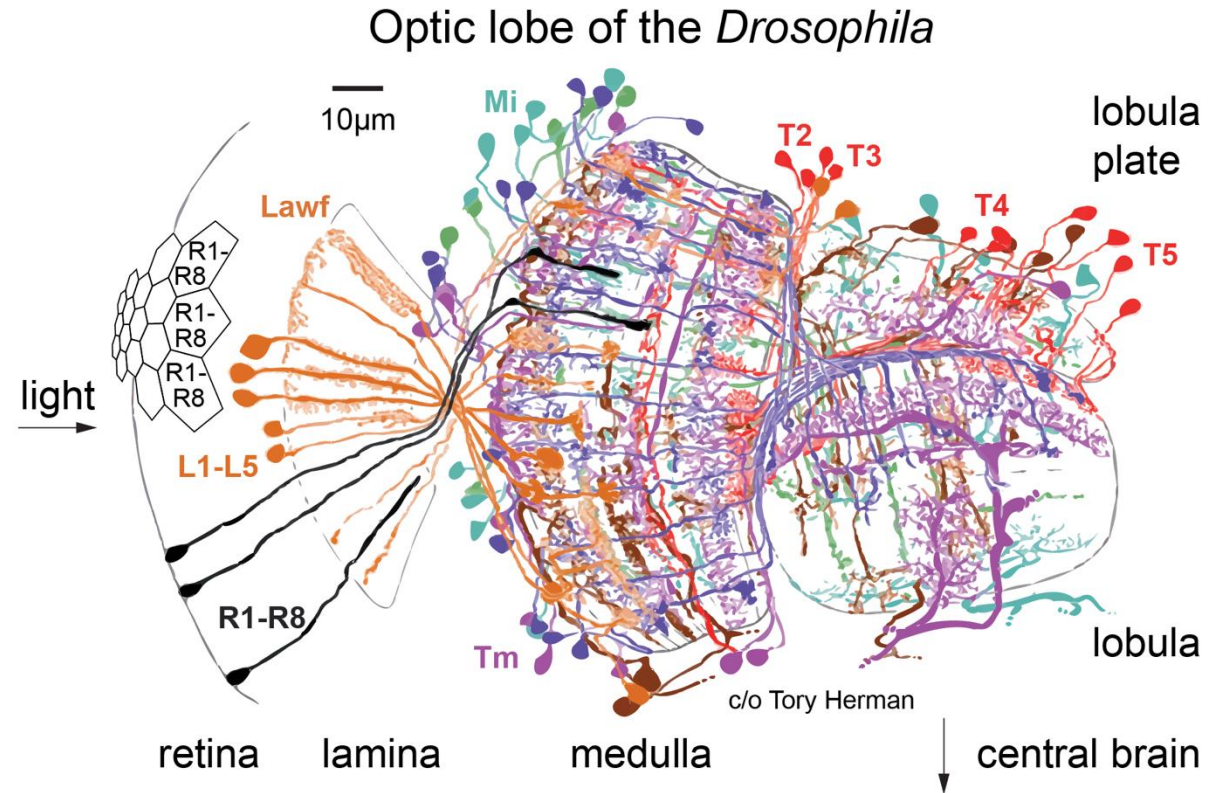
Nern et al. 2024
Rendering by
Siwanowicz I. & Loesche F

My PhD: The fly visual system opportunity

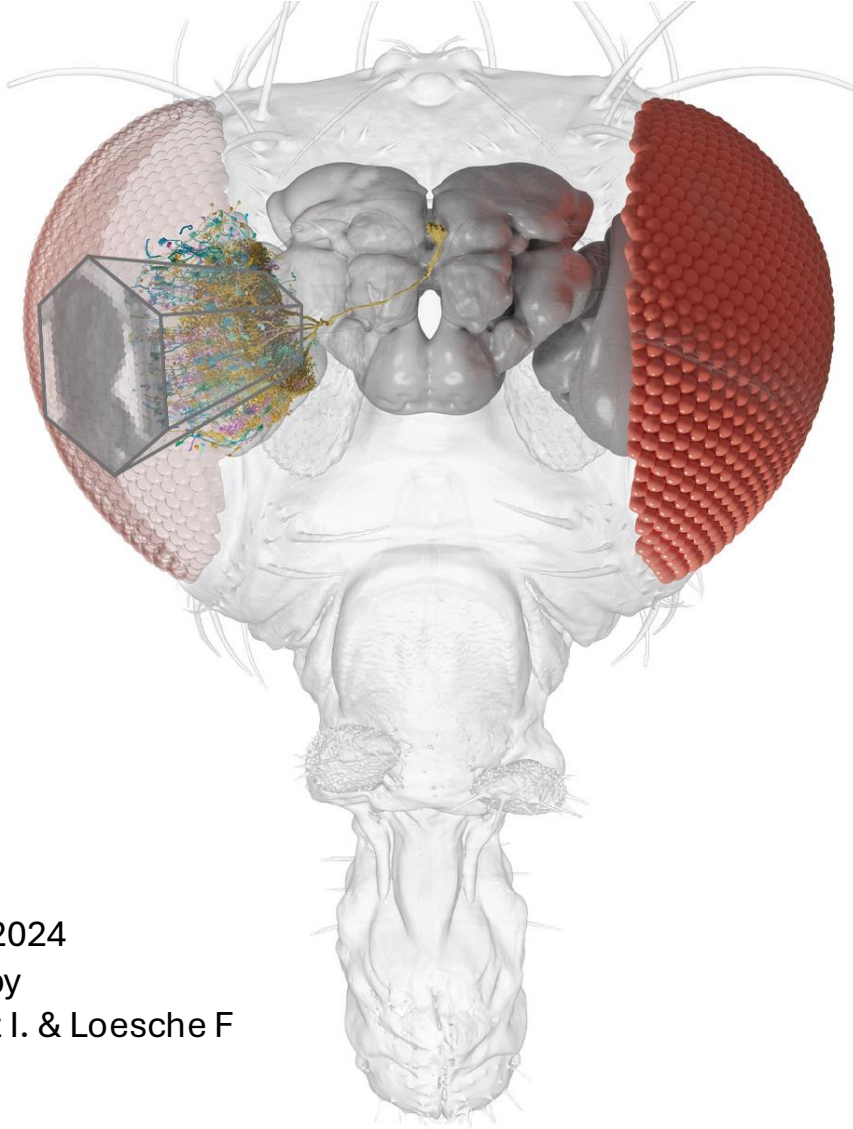


Lappalainen, J. K., Tschopp, F. D., Prakhya, S., McGill, M., Nern, A., Shinomiya, K., Takemura, S.-Y., Gruntman, E., Macke, J. H., & Turaga, S. C. (2024). Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634(8036), 1132–1140.

The fly visual system



Fly visual system is hexagonal and retinotopic



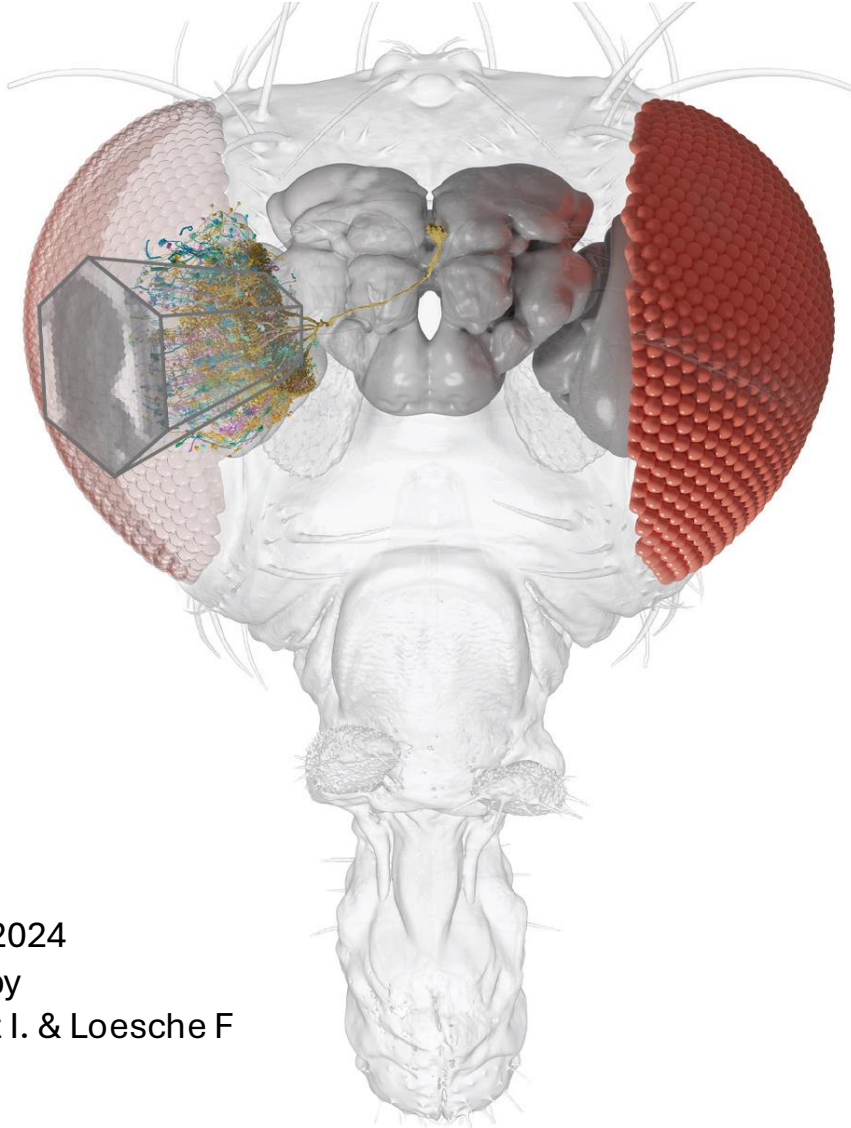
Nern et al. 2024
Rendering by
Siwanowicz I. & Loesche F

Core Results

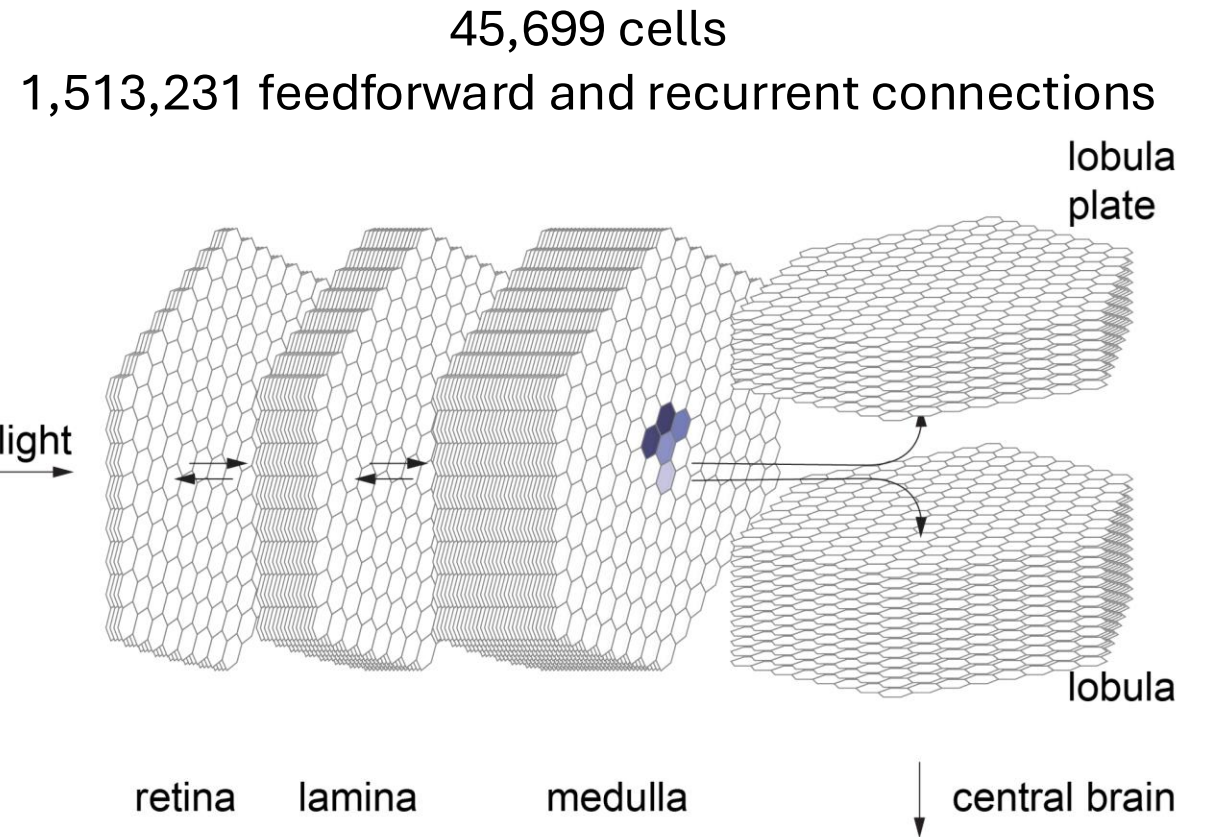
Intro

Summary
Project ideas

We built a retinotopic model of the fly visual system

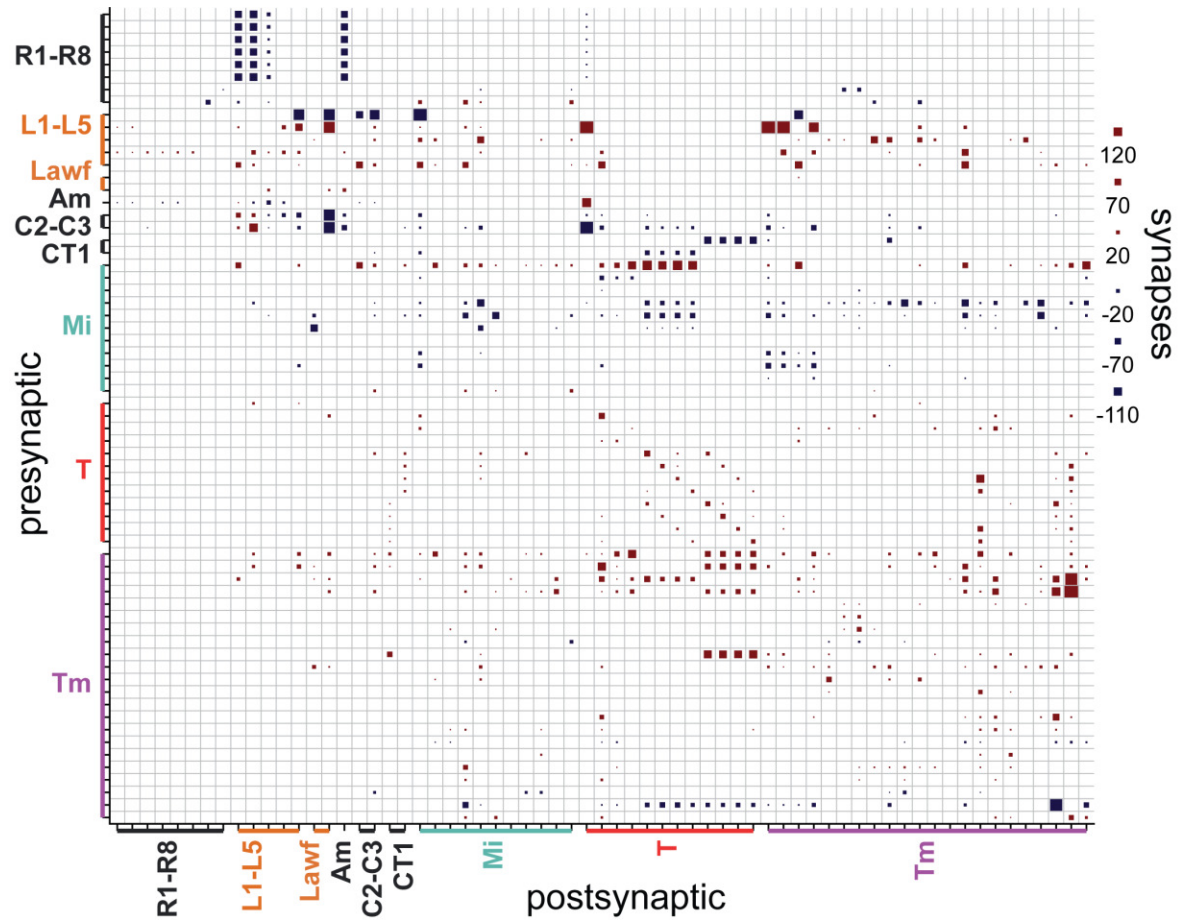


Nern et al. 2024
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Identified connectivity

64 cell types



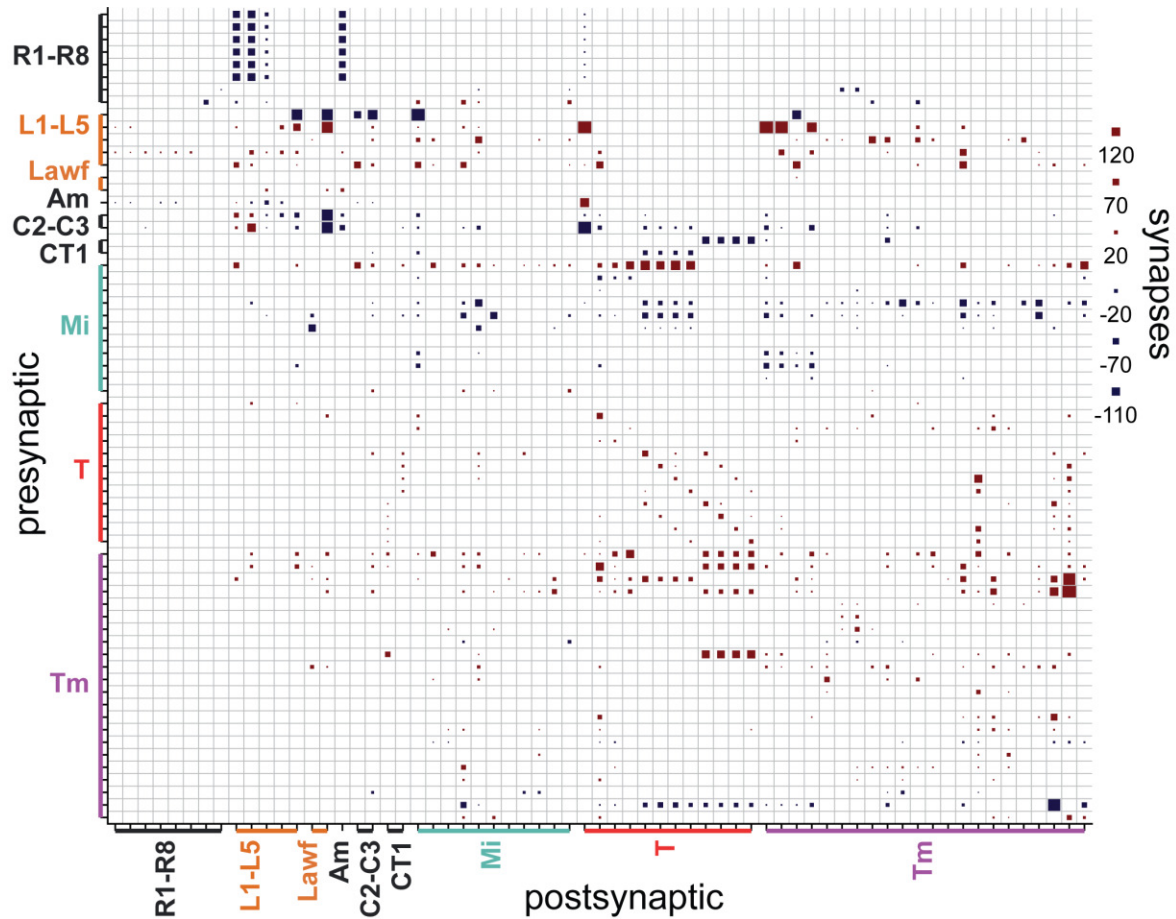
Takemura et al. (2015).
Takemura et al. (2017)
Rivera-Alba et al. (2011)

Tuthill et al. (2014)
Davis et al. (2020).
Hardie (1989)

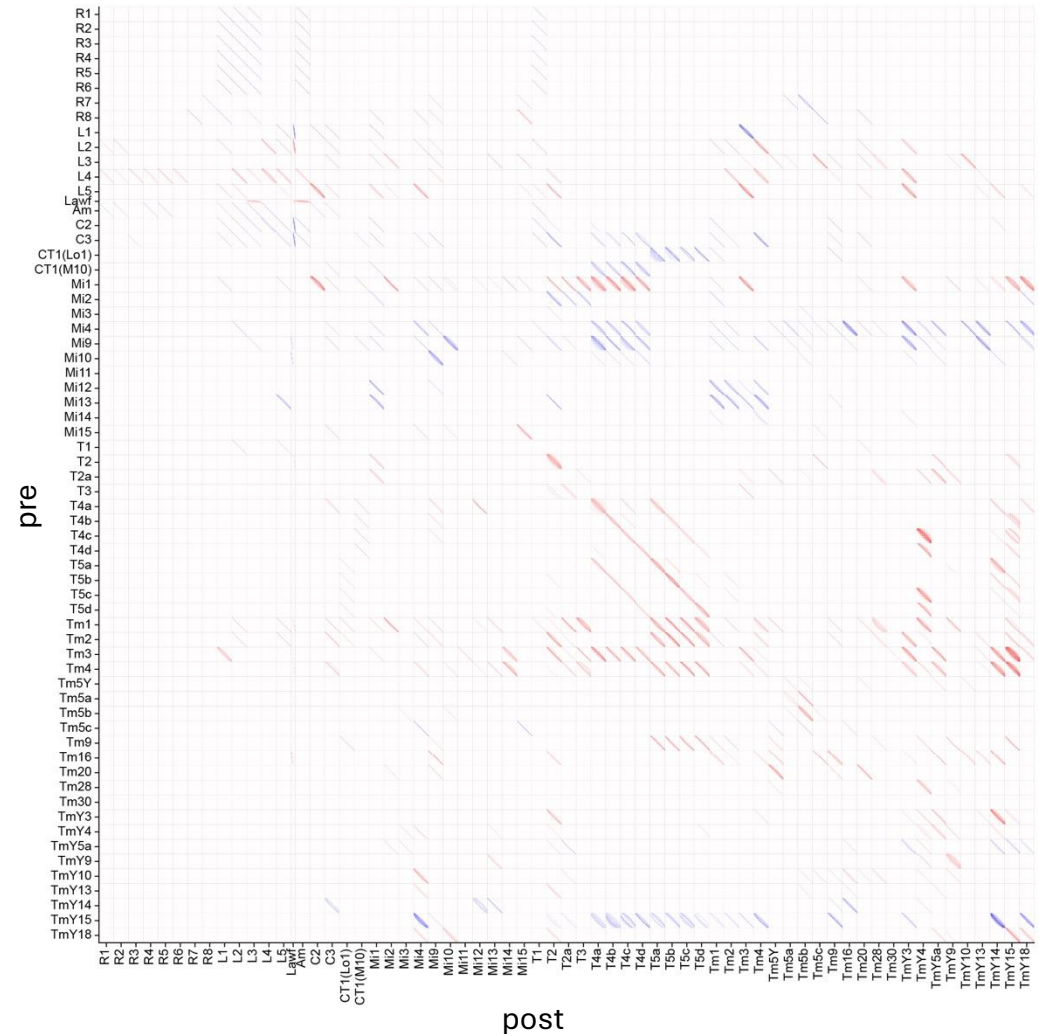
Karuppudurai et al. (2014)
Lin and Potter (2016)
Tschopp et al. (2018)

Identified connectivity

64 cell types



5,759 of 45,699 cells (91 of 721 columns)



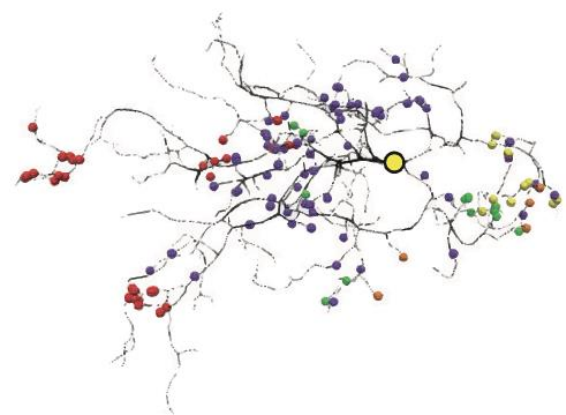
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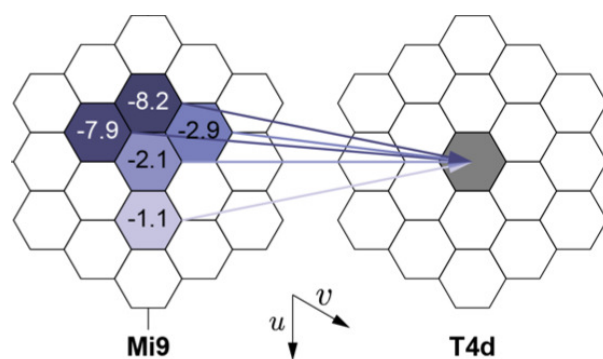
Karuppudurai et al. (2014)
Lin and Potter (2016)
Tschopp et al. (2018)

Connectivity is nearly convolutional

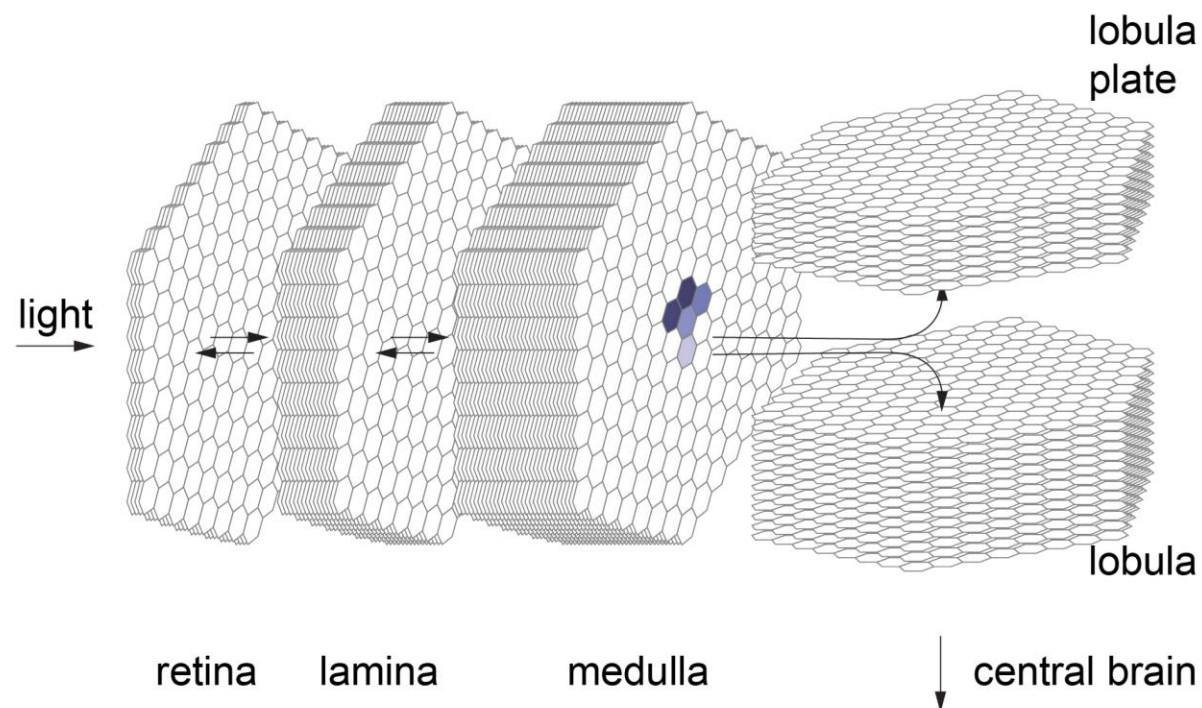
Dendrites with
synapse annotations



Convolutional filter
(anatomical connectivity)



45,699 cells
1,513,231 feedforward and recurrent connections



From synapse counts to a neuron model

weights

$$w_{ij} = \underbrace{\alpha_{t_i t_j}}_{\text{scaling coefficients}} \underbrace{\sigma_{t_i t_j}}_{\text{synapse signs}} \underbrace{N_{t_i t_j, u_i - u_j, v_i - v_j}}_{\text{synapse counts}}$$

From synapse counts to a neuron model

weights

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synapses

$$s_{ij} = w_{ij} \underbrace{f(V_j)}_{\text{transmitter release}}$$


From synapse counts to a neuron model

weights

$$w_{ij} = \underbrace{\alpha_{t_i t_j}}_{\text{scaling coefficients}} \underbrace{\sigma_{t_i t_j} N_{t_i t_j, u_i - u_j, v_i - v_j}}_{\text{synapse signs} \atop \text{synapse counts}}$$

synapses

$$s_{ij} = w_{ij} f(V_j)$$

|
transmitter release



voltage

$$\underbrace{\tau_{t_i}}_{\text{time constants}} \frac{dV_i}{dt} = -V_i + \sum_j \underbrace{s_{ij}}_{\text{resting potentials}} + \underbrace{V_{t_i}^{\text{rest}}}_{\text{resting potentials}} + e_i$$

From synapse counts to a neuron model

weights

$$w_{ij} = \underbrace{\alpha_{t_i t_j}}_{\text{scaling coefficients}} \underbrace{\sigma_{t_i t_j}}_{\text{synapse signs}} \underbrace{N_{t_i t_j, u_i - u_j, v_i - v_j}}_{\text{synapse counts}}$$

synapses

$$s_{ij} = w_{ij} f(V_j)$$

|
transmitter release



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$$\underbrace{\tau_{t_i}}_{\text{time constants}} \frac{dV_i}{dt} = -V_i + \sum_j \underbrace{s_{ij}}_{\text{resting potentials}} + \underbrace{V_{t_i}^{\text{rest}}}_{\text{resting potentials}} + e_i$$

Parameter efficiency

Connectome: 734 free parameters

Comparison:

- Without cell connectivity: 2B
- or ~400K when assuming local convolutions
- AlexNet: 60M

End-to-end equations

Rendering

$$\{\mathbf{X}, \mathbf{Y}\} = \text{FlyEyeRender}(\text{Sintel})$$

DMN

$$\tau_{t_i} \frac{dV_i}{dt} = -V_i + \sum_j \alpha_{t_i t_j} \sigma_{t_i t_j} N_{t_i t_j} \Delta u \Delta v f(V_j) + V_{t_i}^{\text{rest}} + x_{u_i v_i}$$

Decoding

$$\hat{\mathbf{Y}}_t = \text{Decoder}(\text{DMN}(\mathbf{X}_0, \dots, \mathbf{X}_t))$$

Initialization

$$\alpha_{t_i, t_j} = \frac{0.01}{\langle N_{t_i, t_j} \rangle_{\Delta u, \Delta v}}$$

$$V_{t_i}^{\text{rest}} \sim \mathcal{N}(0.5, 0.05)$$

$$\tau_{t_i} = 0.05$$

Task-training

$$\tau_{t_i}^*, \alpha_{t_i t_j}^* = \arg \min_{\tau_{t_i}, \alpha_{t_i t_j}} \mathcal{L}_{\text{task}}$$

$$V_{t_i}^{\text{rest},*} = \arg \min_{V_{t_i}^{\text{rest}}} \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{reg}}$$

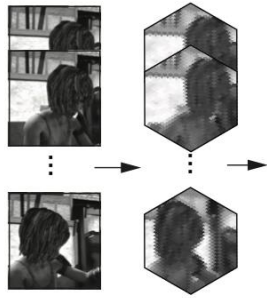
where:

$$\mathcal{L}_{\text{task}} = \|\hat{\mathbf{Y}} - \mathbf{Y}\|$$

$$\mathcal{L}_{\text{reg}} = \frac{\lambda_V}{T} \sum_{t_{\text{central}}} \begin{cases} \gamma(\bar{V} - a)^2, & \text{if } \bar{V} = \frac{1}{N} \sum_n V_{t_{\text{central}}}[n] \leq a \\ \delta(\bar{V} - a)^2, & \text{if } \bar{V} > a \end{cases}$$

We task-train the unknown neural parameters

video

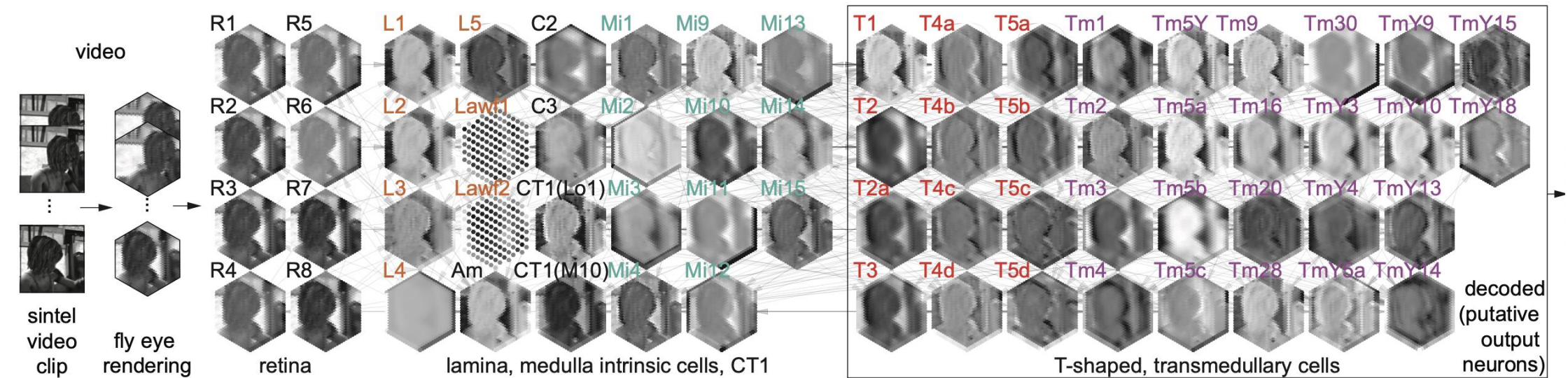


sintel
video
clip

fly eye
rendering

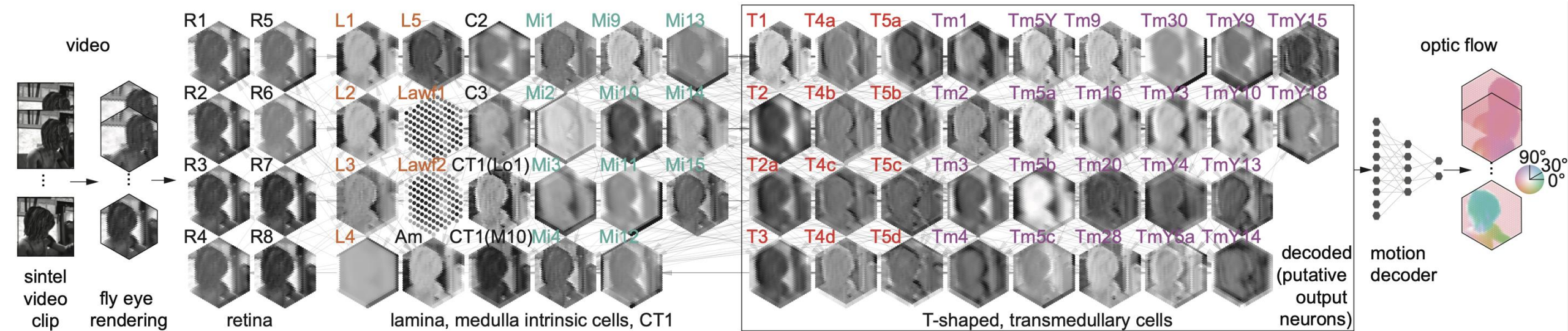
Butler, et al.
(2012).

We task-train the unknown neural parameters



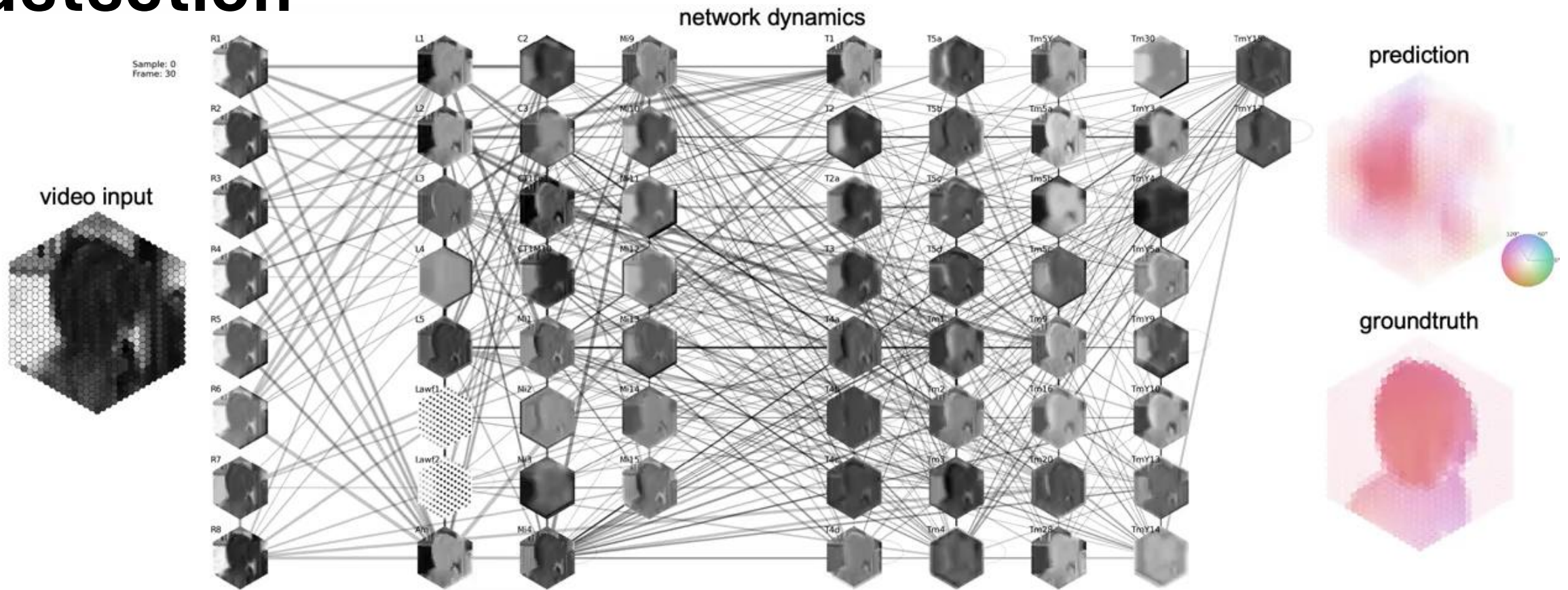
Butler, et al.
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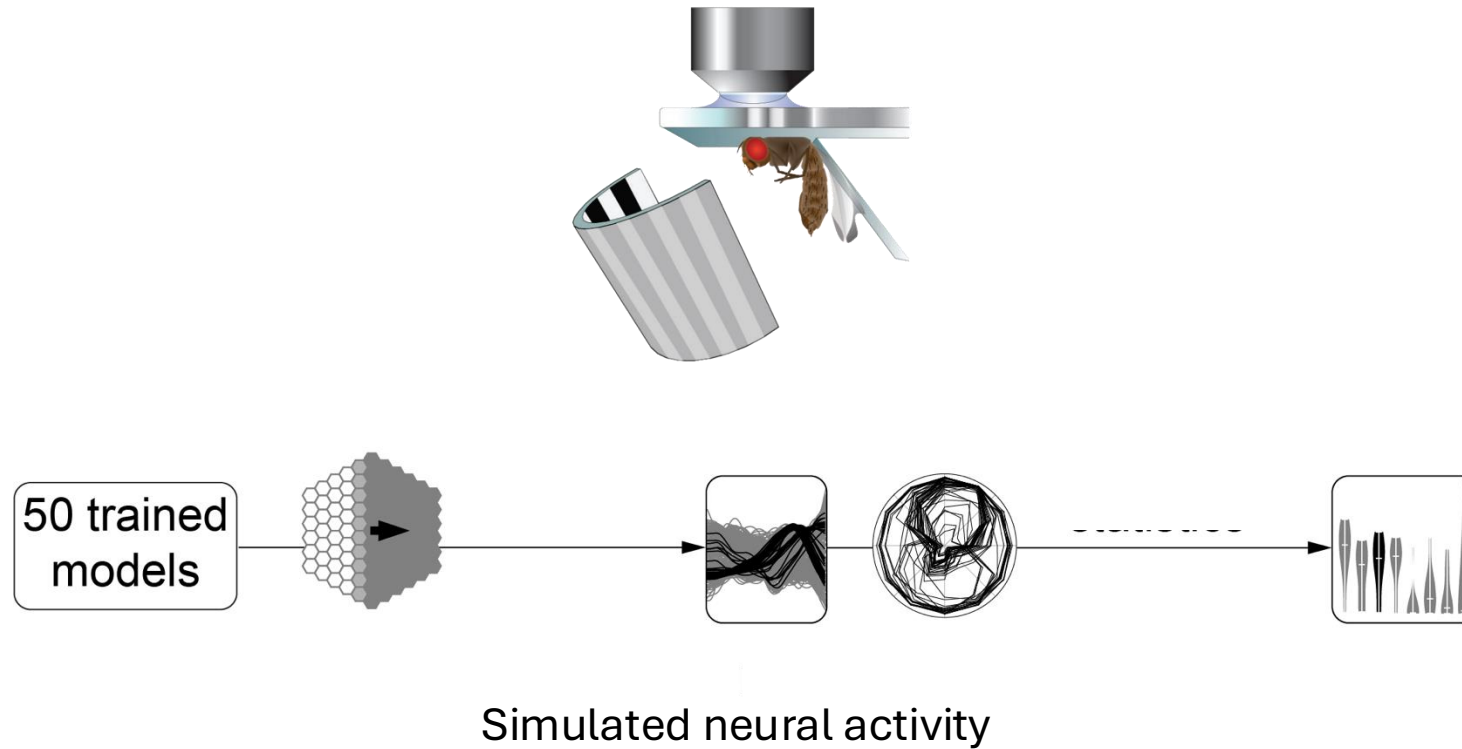


Butler, et al.
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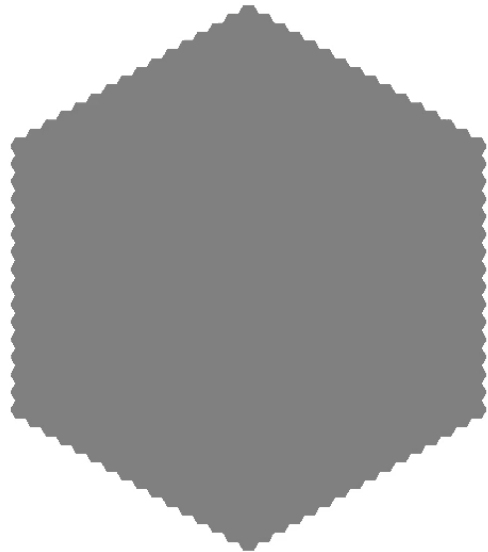
Task-trained network dynamics and motion detection



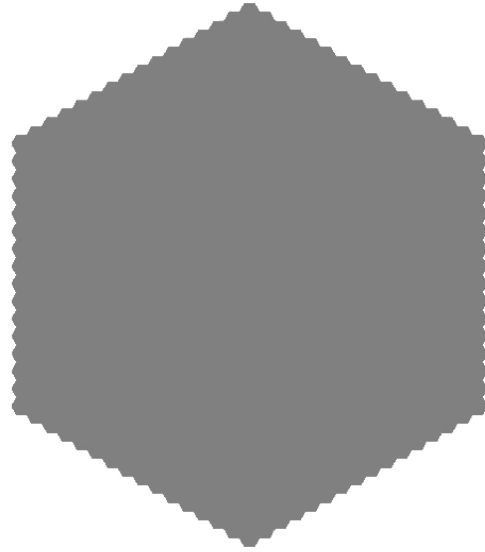
Validating trained models



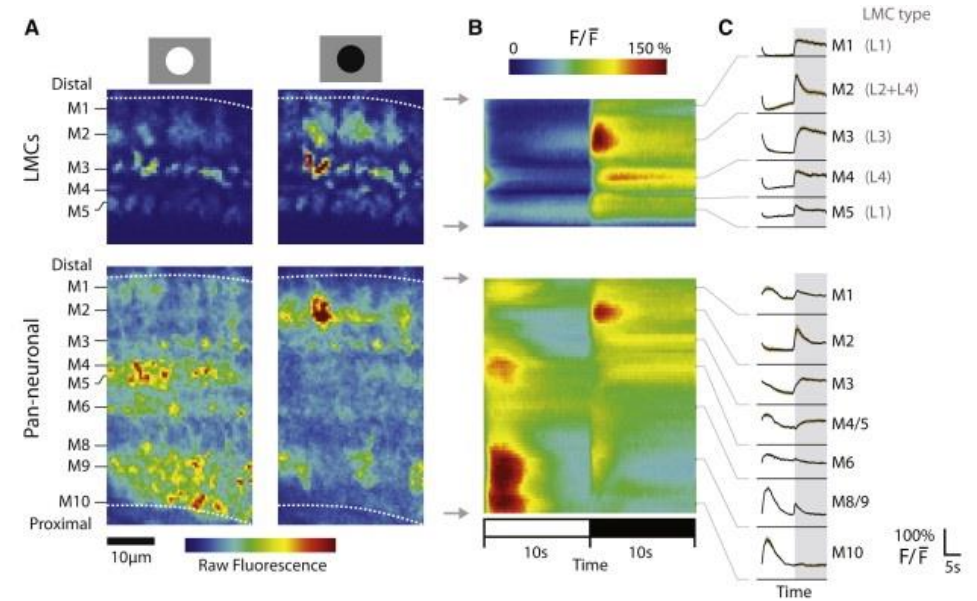
Segregation into ON and OFF pathways



ON flash



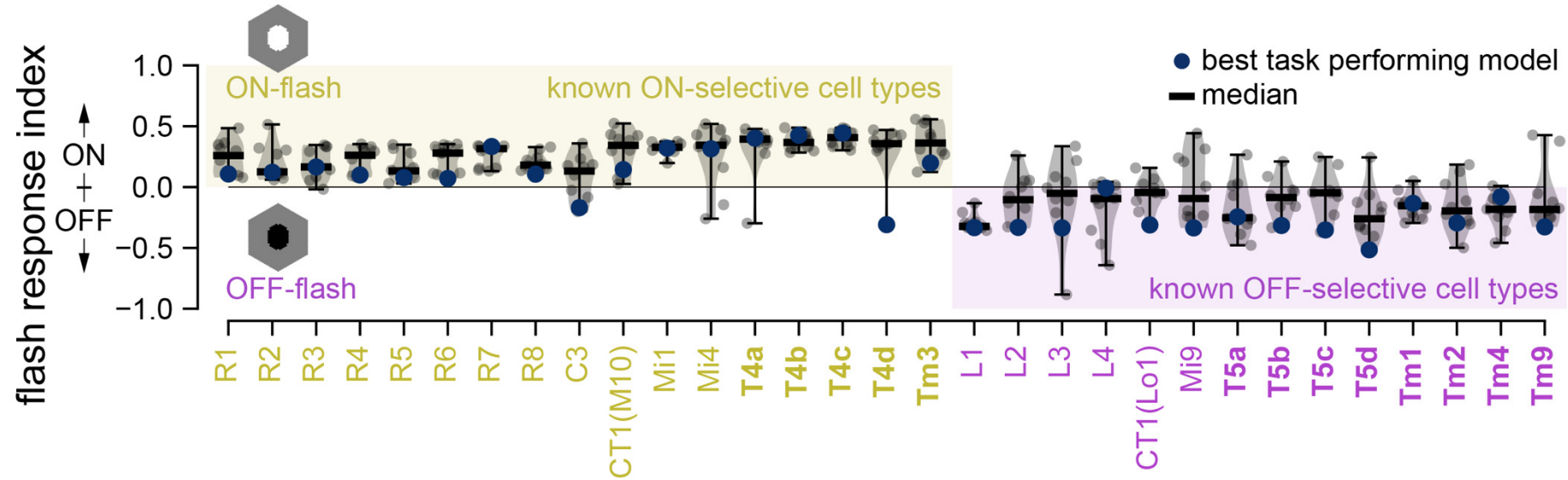
OFF flash



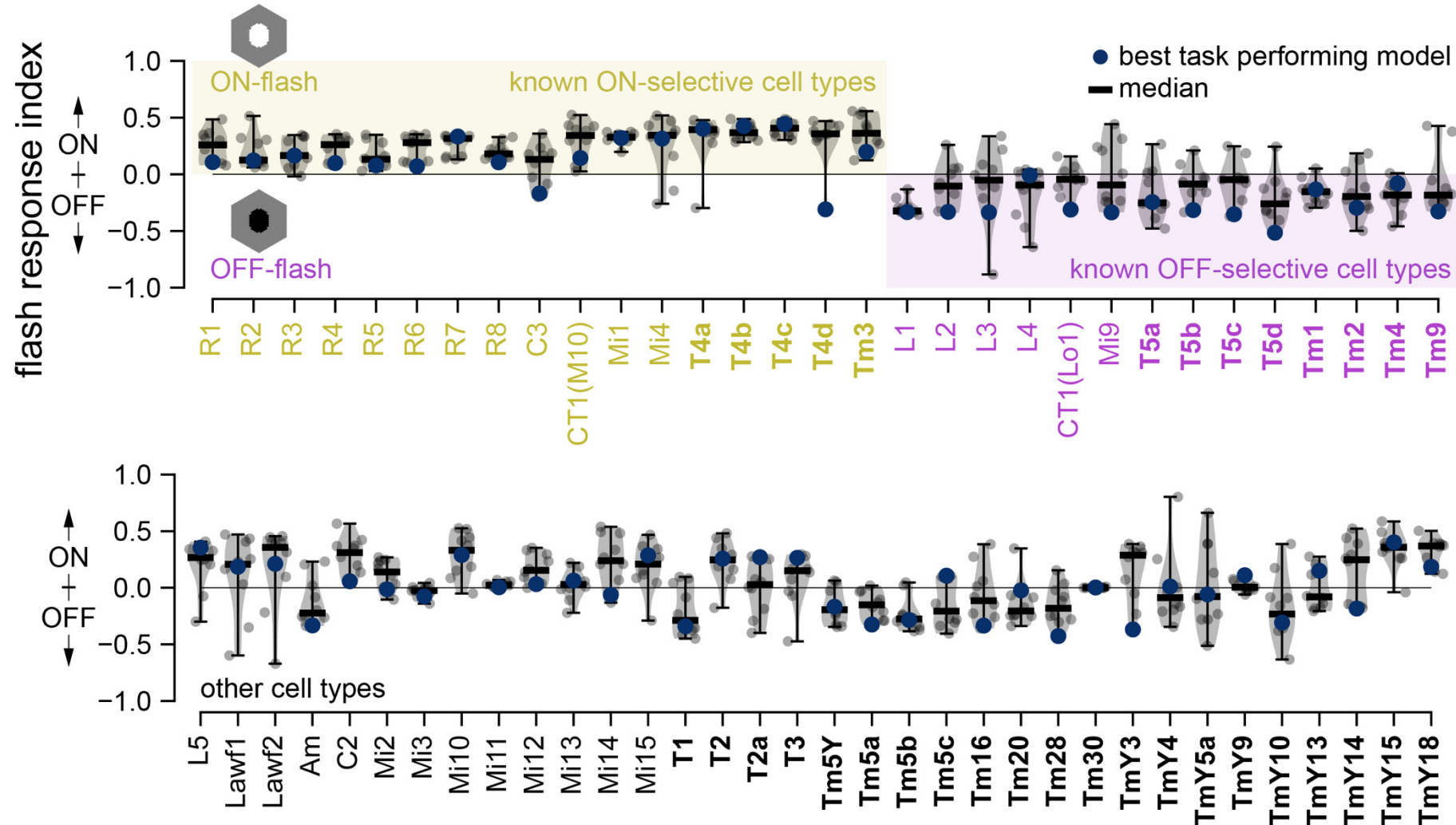
Strother et al. (2014).

Direct Observation of ON and OFF Pathways in the Drosophila Visual System.

Model predicts ON and OFF pathways

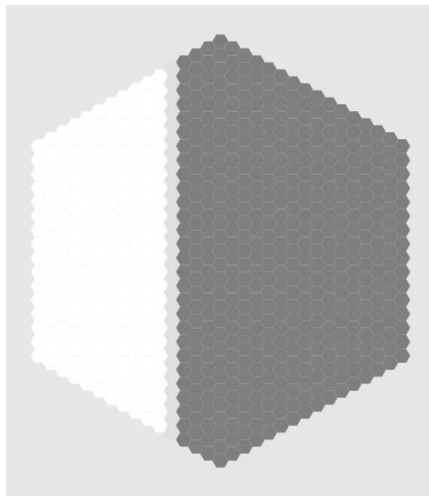


Model predicts ON and OFF pathways

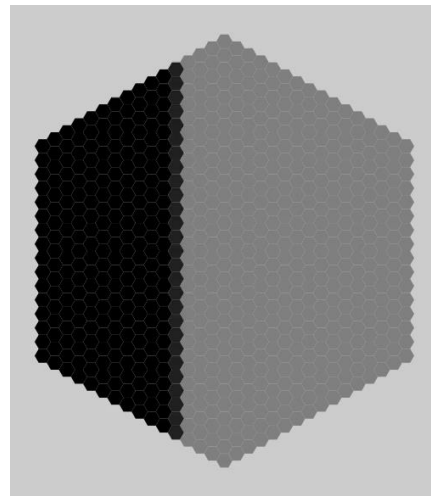


Novel
hypotheses

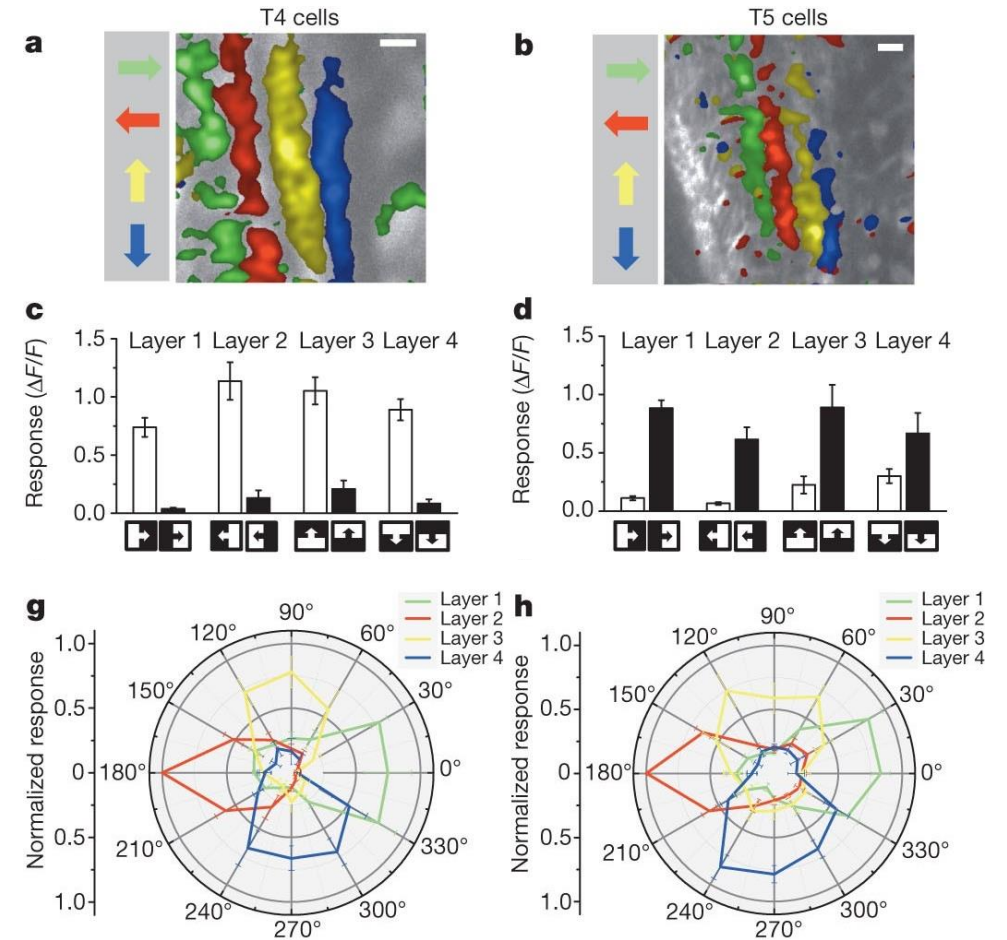
Motion vision mechanisms



ON moving edge



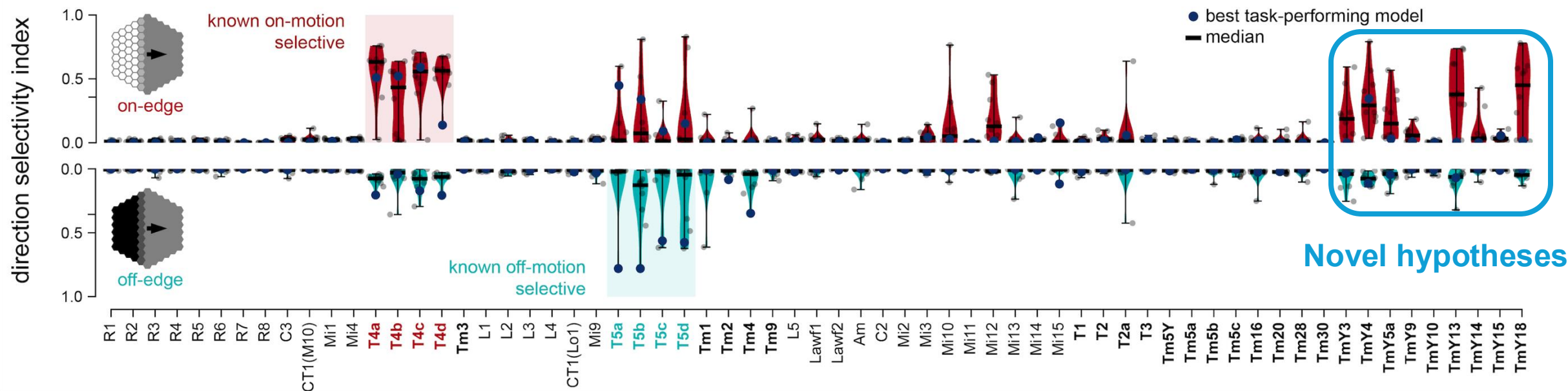
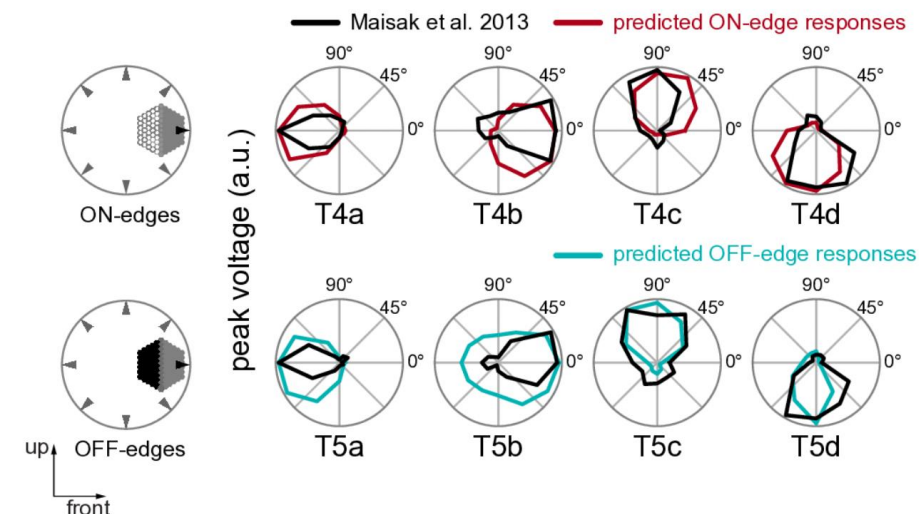
OFF moving edge



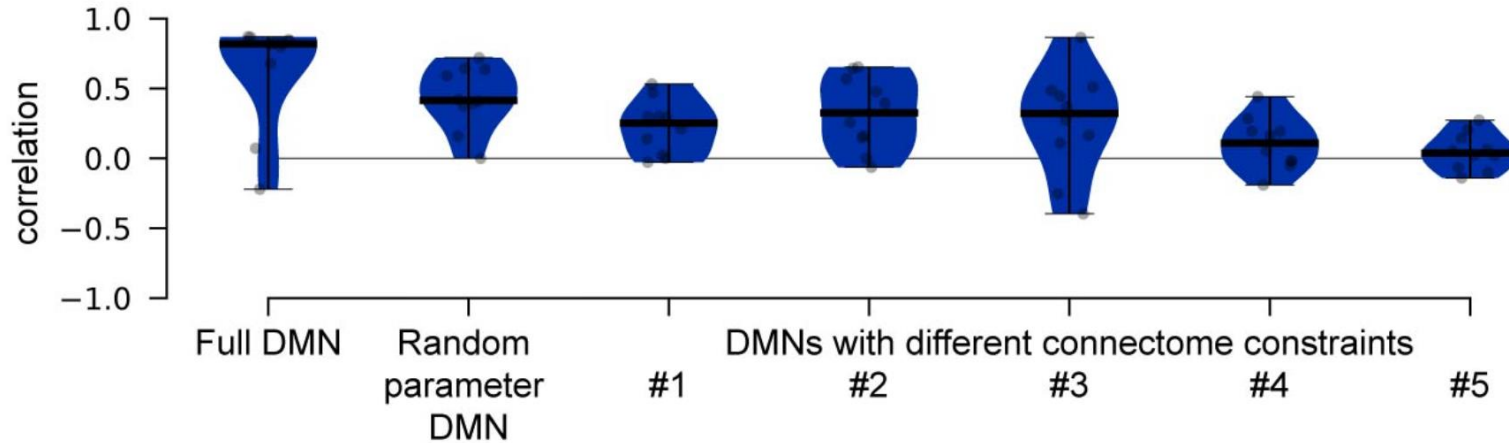
Maisak et al. (2013).

A directional tuning map of Drosophila elementary motion detectors.

Model predicts motion vision mechanisms

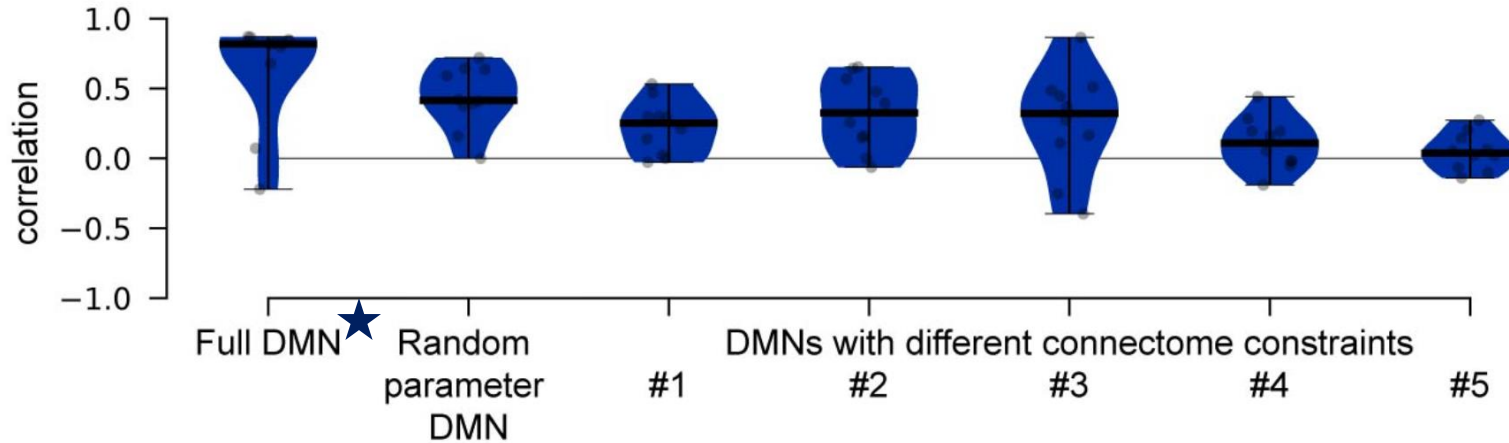


Models with full connectomic constraints and task-optimization predict neural function best



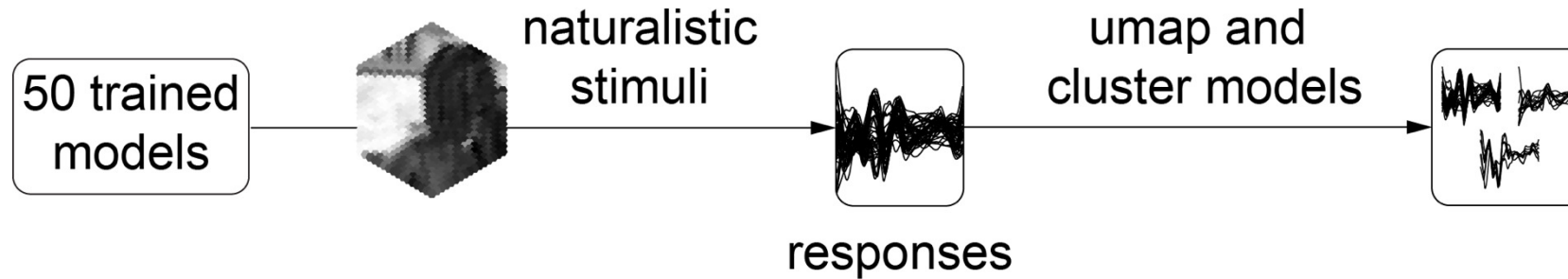
#	Synapse signs	Single-neuron connectivity	Synapse counts	Task-trained constraint(s)	Weight expression
1	Known ($\sigma_{t_i t_j}$)	Known ($\Delta u, \Delta v$)	Unknown	Synapse counts and unitary scaling	$w_{ij} = \sigma_{t_i t_j} \mathbf{m}_{t_i t_j} \Delta u, \Delta v$
2	Known	Unknown	Unknown	Average convolutional filter*	$w_{ij} = \sigma_{t_i t_j} \mathbf{m}_{t_i t_j, -3 < \Delta u, \Delta v, \Delta w < 3}$
3	Unknown	Known	Known ($N_{...}$)	Synapse sign (Dale's principle) and unitary scaling	$w_{ij} = \alpha_{t_i t_j} \sigma_{t_j} N_{t_i t_j, \Delta u, \Delta v}$
4	Unknown	Known	Unknown	All weights	$w_{ij} = \mathbf{w}_{t_i t_j} \Delta u \Delta v$
5	Unknown	Unknown	Unknown	All weights	$w_{ij} = \mathbf{w}_{t_i t_j, -3 < \Delta u, \Delta v, \Delta w < 3}$

Models with full connectomic constraints and task-optimization predict neural function best

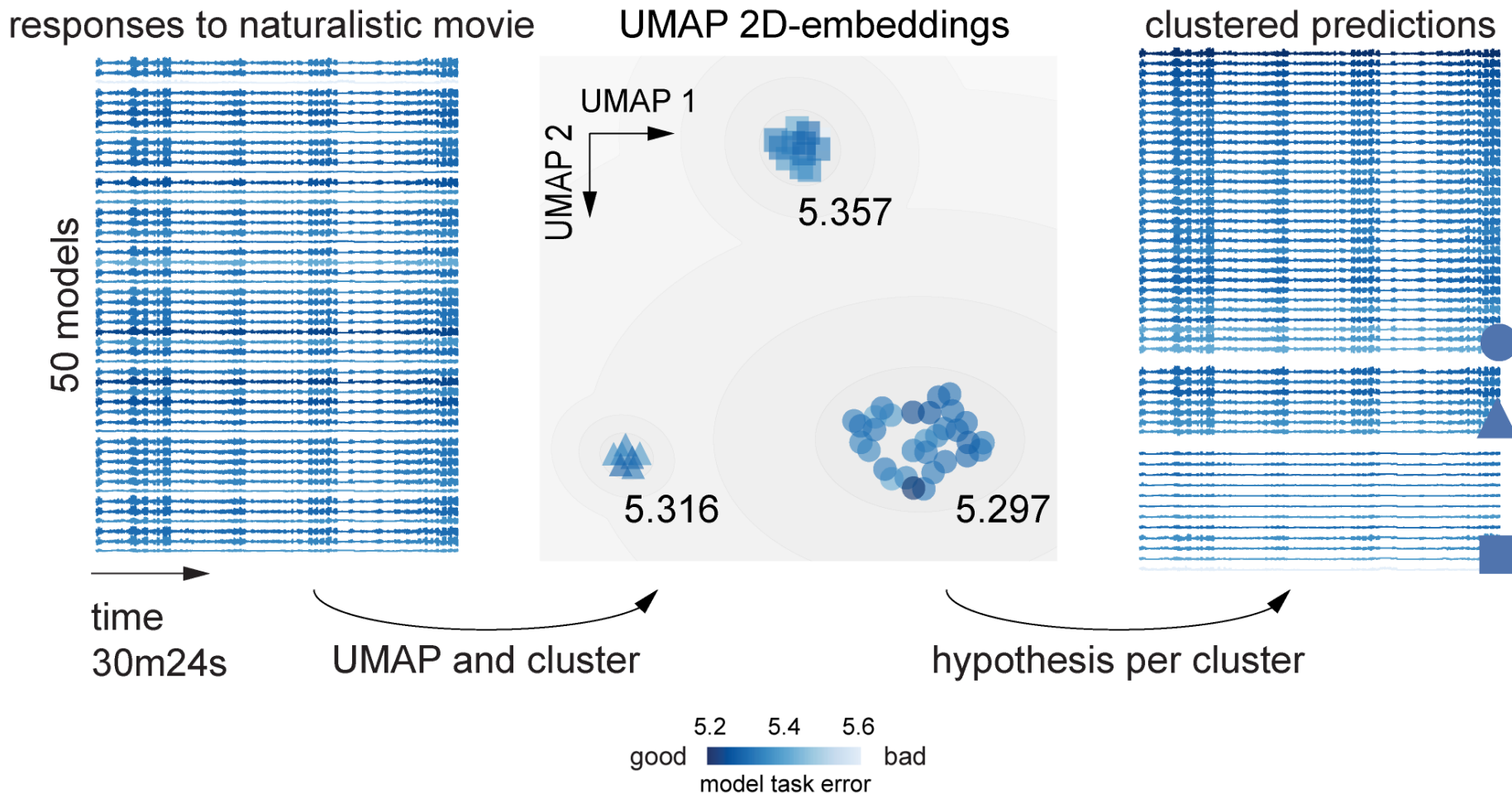


#	Synapse signs	Single-neuron connectivity	Synapse counts	Task-trained constraint(s)	Weight expression
1	Known ($\sigma_{t_i t_j}$)	Known ($\Delta u, \Delta v$)	Unknown	Synapse counts and unitary scaling	$w_{ij} = \sigma_{t_i t_j} \mathbf{m}_{t_i t_j} \Delta u, \Delta v$
2	Known	Unknown	Unknown	Average convolutional filter*	$w_{ij} = \sigma_{t_i t_j} \mathbf{m}_{t_i t_j, -3 < \Delta u, \Delta v, \Delta w < 3}$
3	Unknown	Known	Known ($N_{...}$)	Synapse sign (Dale's principle) and unitary scaling	$w_{ij} = \alpha_{t_i t_j} \sigma_{t_j} N_{t_i t_j, \Delta u, \Delta v}$
4	Unknown	Known	Unknown	All weights	$w_{ij} = \mathbf{w}_{t_i t_j} \Delta u \Delta v$
5	Unknown	Unknown	Unknown	All weights	$w_{ij} = \mathbf{w}_{t_i t_j, -3 < \Delta u, \Delta v, \Delta w < 3}$

Understanding variable predictions across trained models

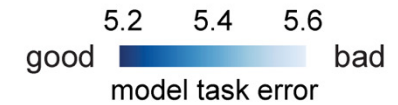
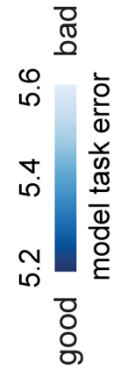
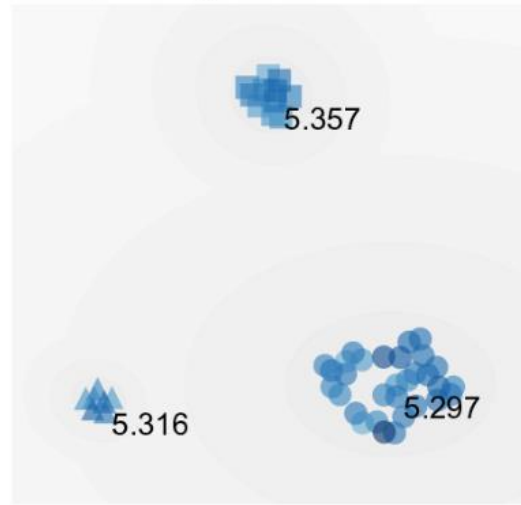


Nonlinear dimensionality reduction and clustering



Model ensemble clusters into hypotheses

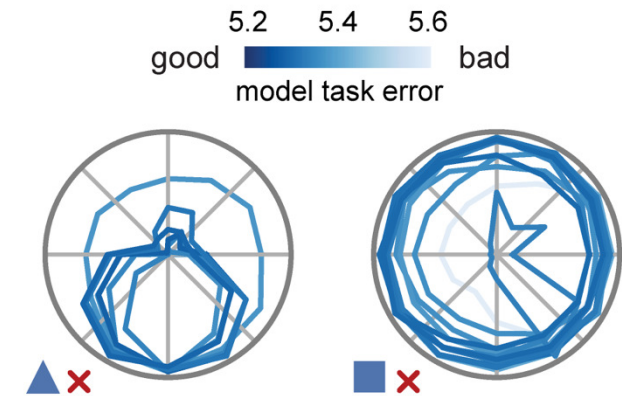
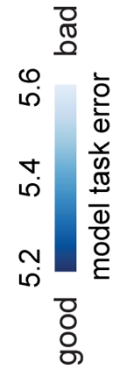
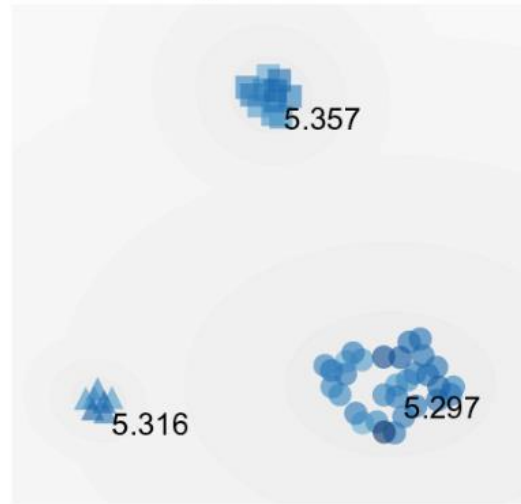
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Example cell type: T4c

Model ensemble clusters into hypotheses

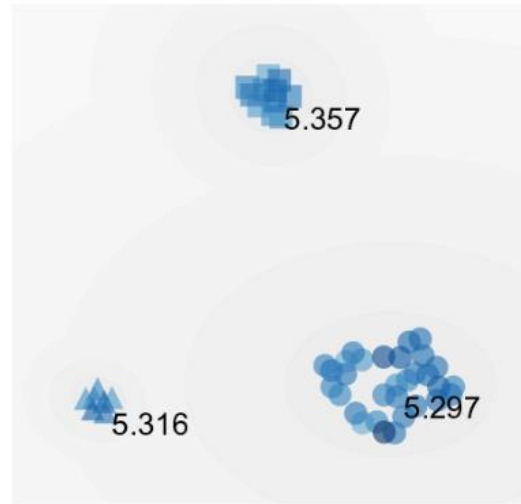
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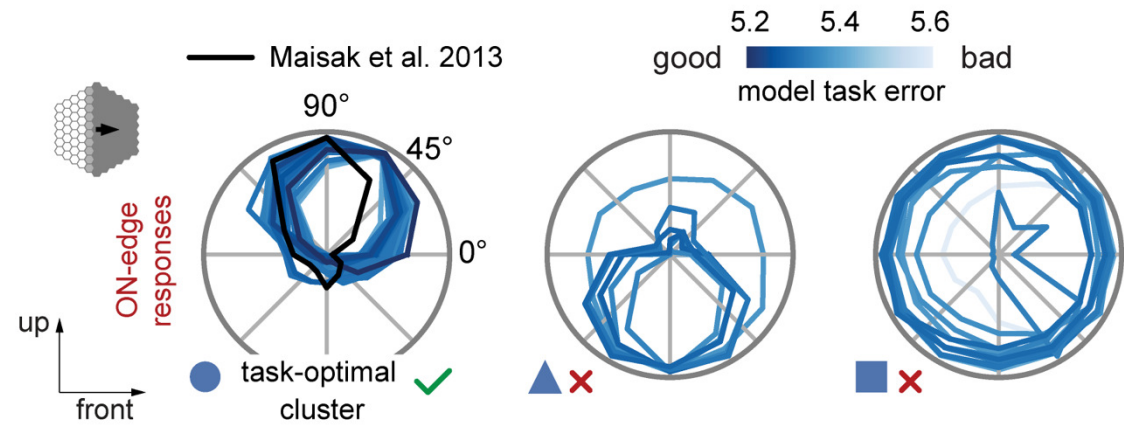
Example cell type: T4c

Model ensemble clusters into hypotheses

Example cell type: T4c



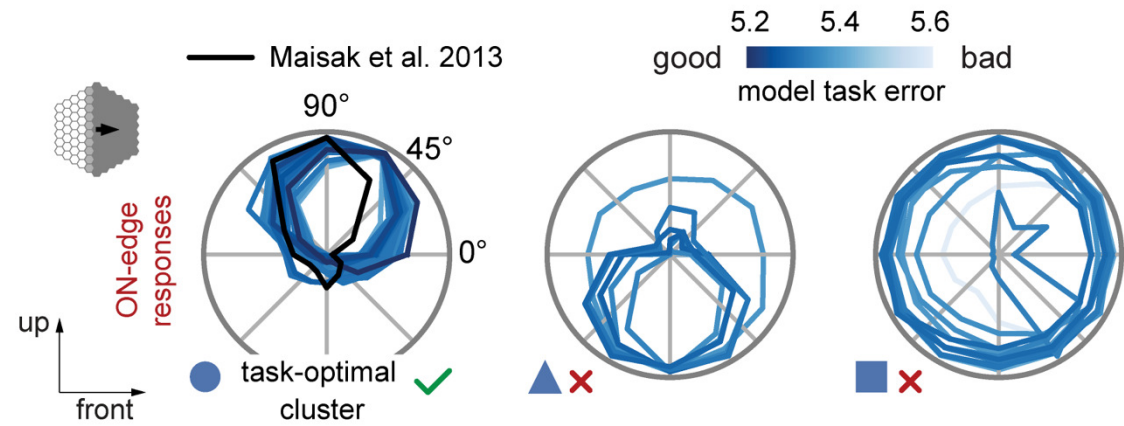
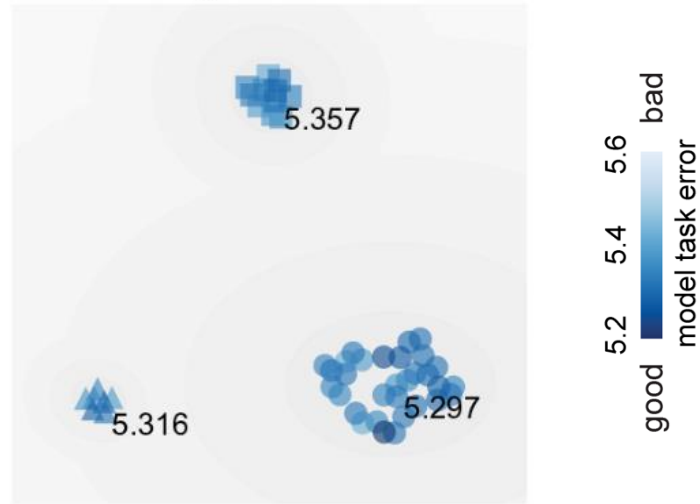
5.2 5.4 5.6
good bad
model task error



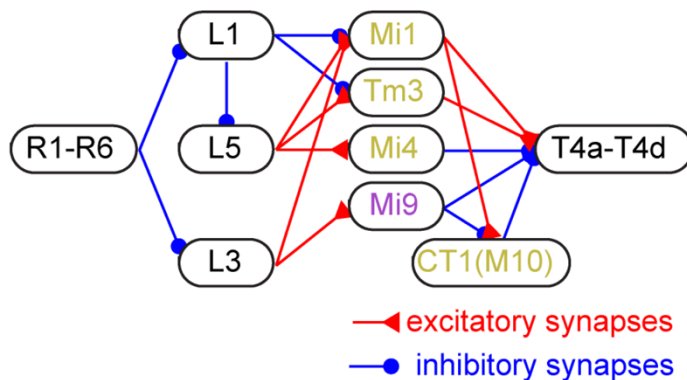
Example cell type: T4c

Activity measurements to refine model

Example cell type: T4c

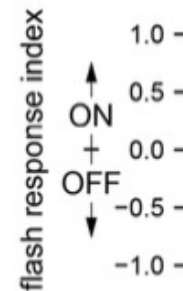
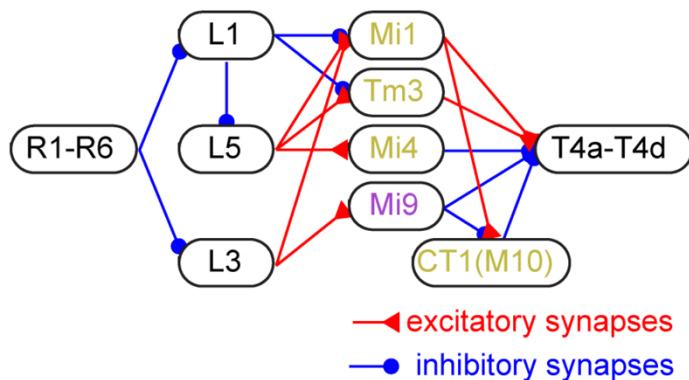
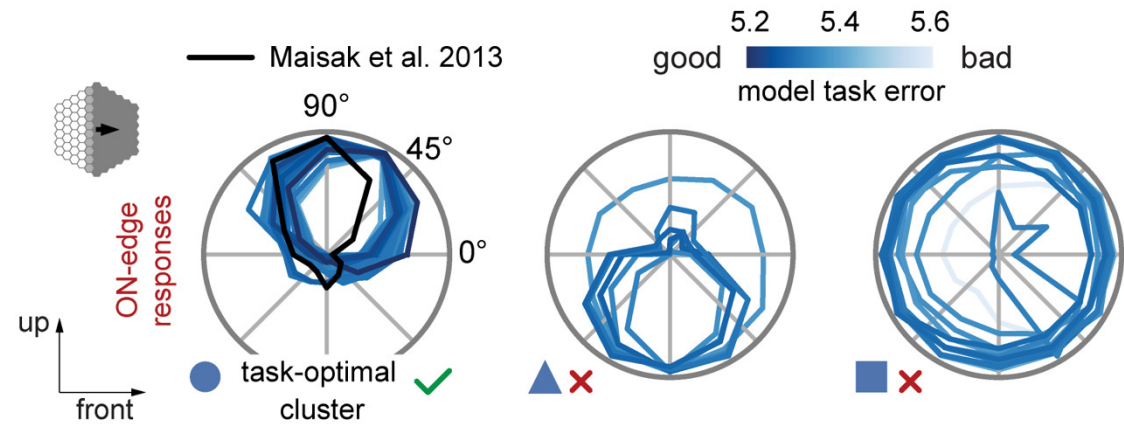
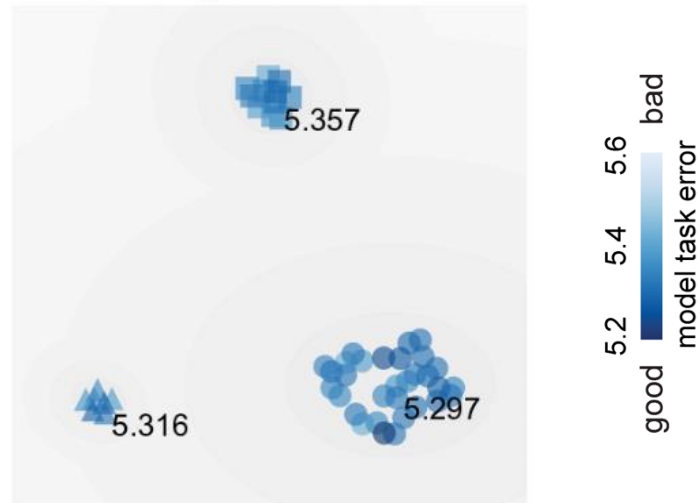


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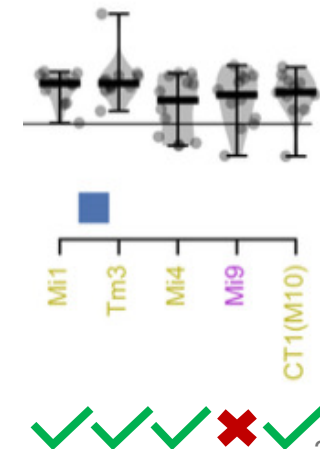
Activity measurements to refine model

Example cell type: T4c



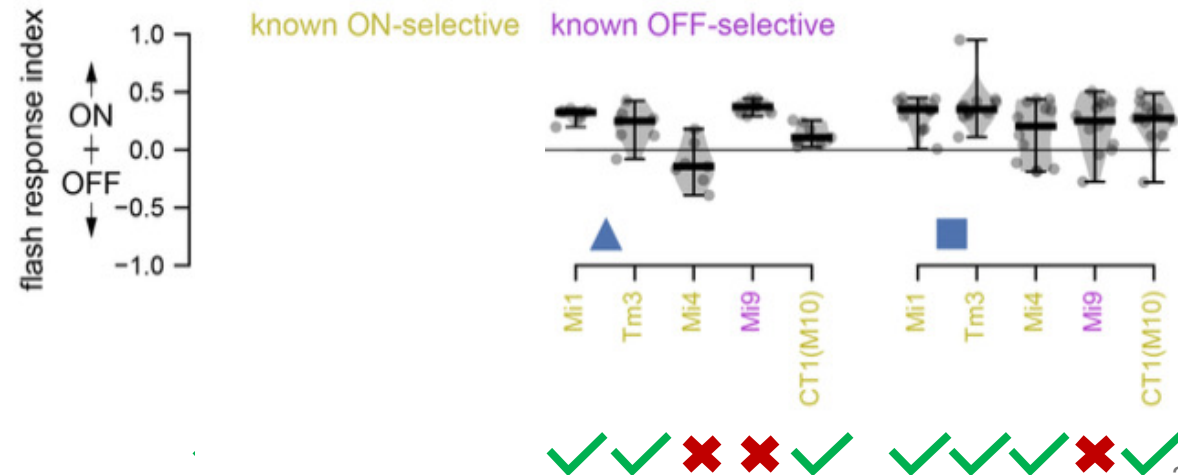
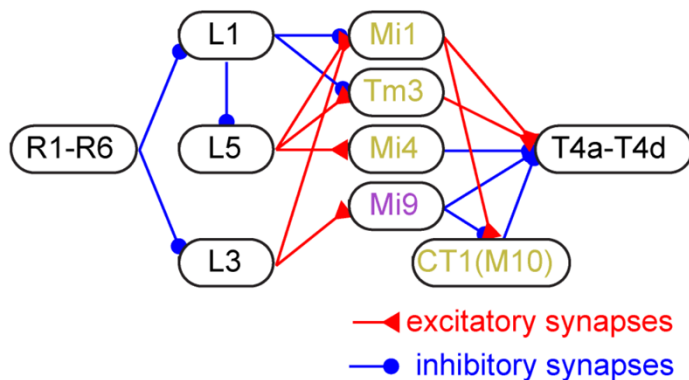
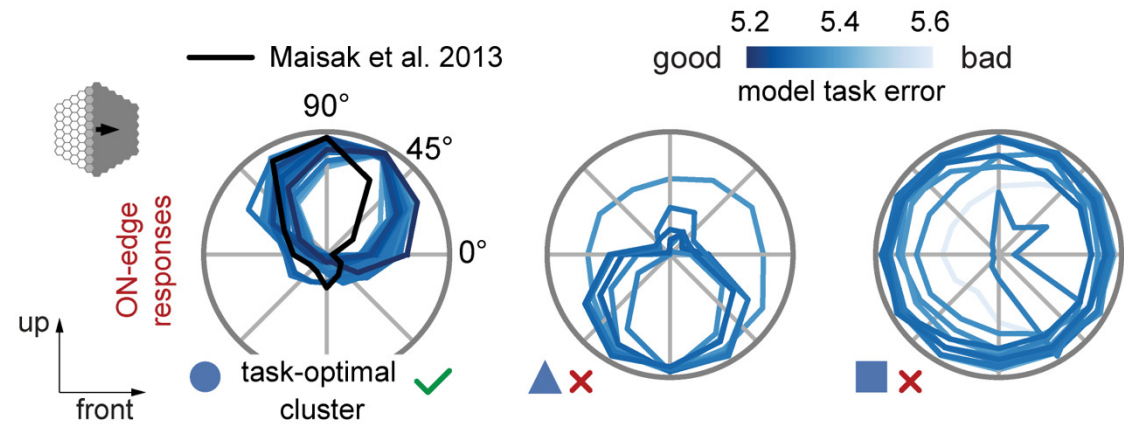
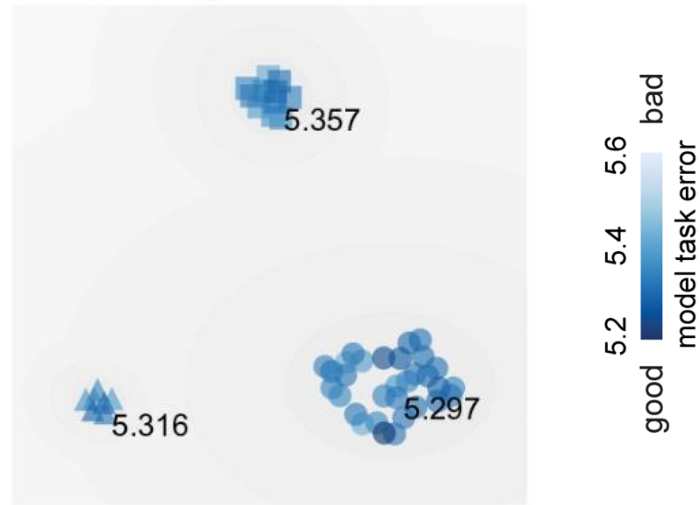
known ON-selective known OFF-selective

Example cell type: T4c



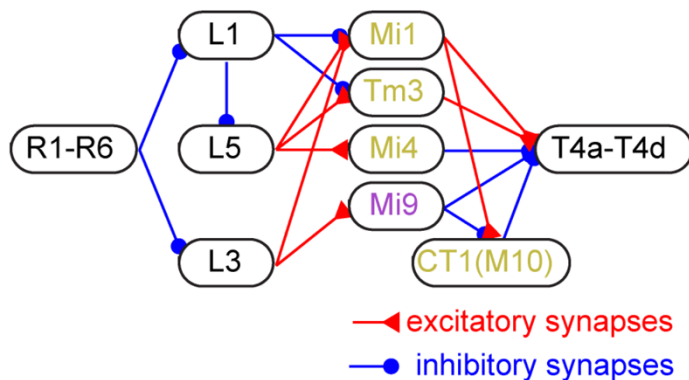
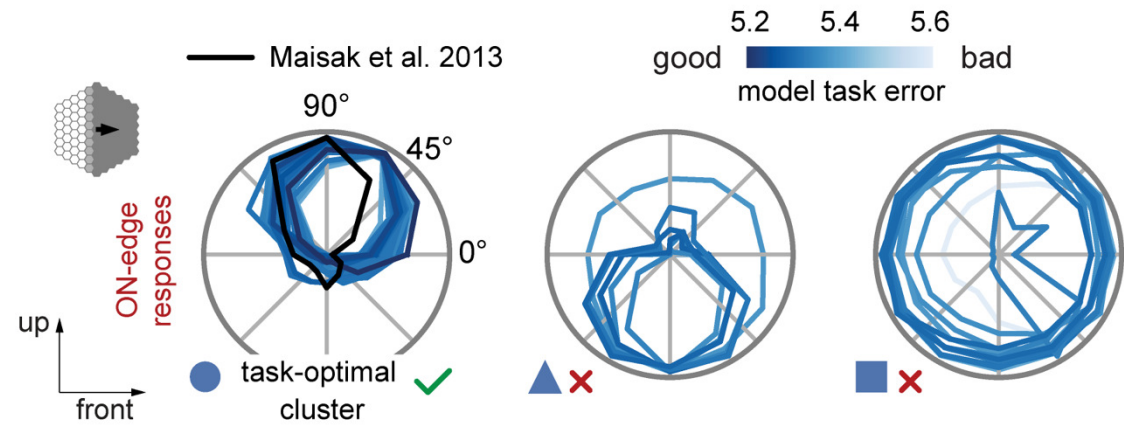
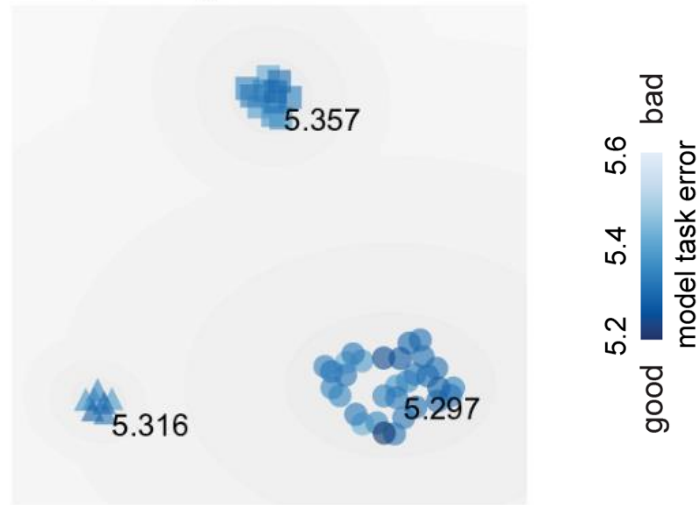
Activity measurements to refine model

Example cell type: T4c

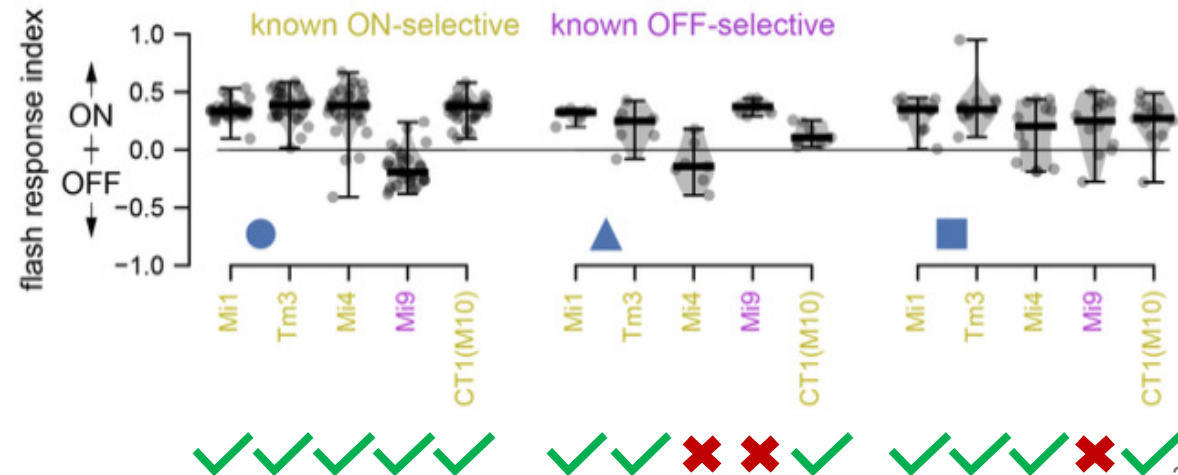


Activity measurements to refine model

Example cell type: T4c



Example cell type: T4c



Summary

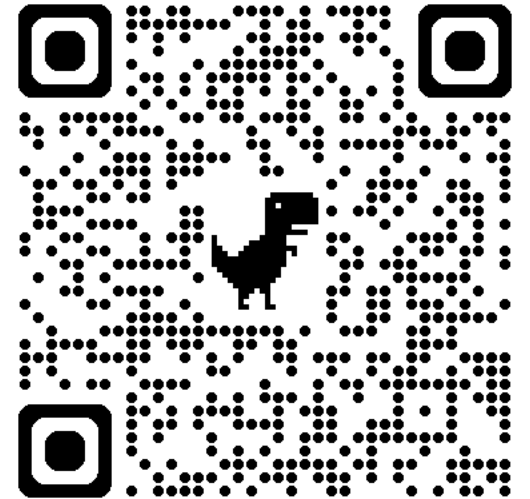
Core Results

Intro

Project ideas

Summary: From connectome to computation with machine learning

- Using the connectome and task-optimization, we built a dynamic model of the fly visual system that computes motion.
- We validated the model across experimental measurements from 26 studies spanning decades of experimental research.
- Model hypotheses can be refined with experimental measurements.
- Whole nervous system computation remains fundamentally challenging to explain in many details.
- But we have a general strategy now for building and interpreting dynamical models from connectomes:
- Parametrizing unknowns, optimizing via tasks, and distilling hypotheses from ensembles with clustering and measurements of neural activity.



Link to publication.



Fabian
Tschopp



Dr.
Aljoscha
Nern



Sridhama
Prakya



Dr. Eyal
Gruntman



Turagalab



Mackelab

**Thanks for feedback and
discussions**

Lou Scheffer
Lowell Umayam
Axel Borst
James Fitzgerald
Nathan Klapoetke

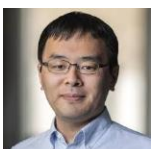
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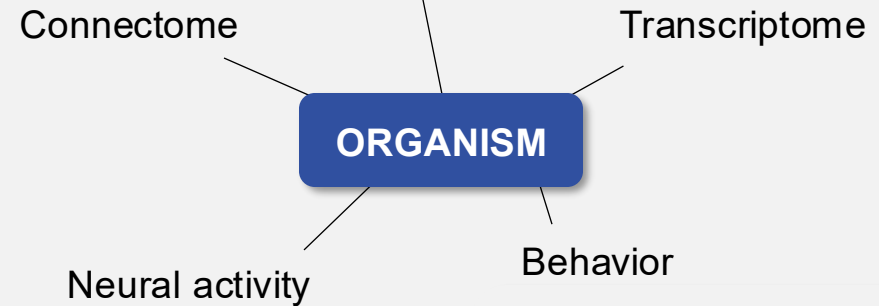
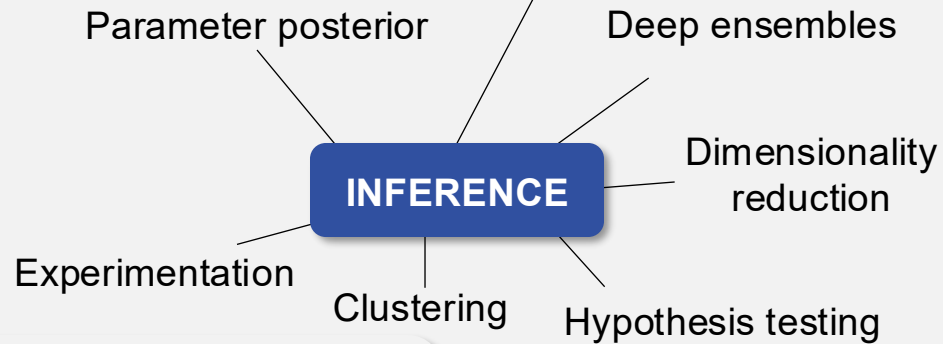
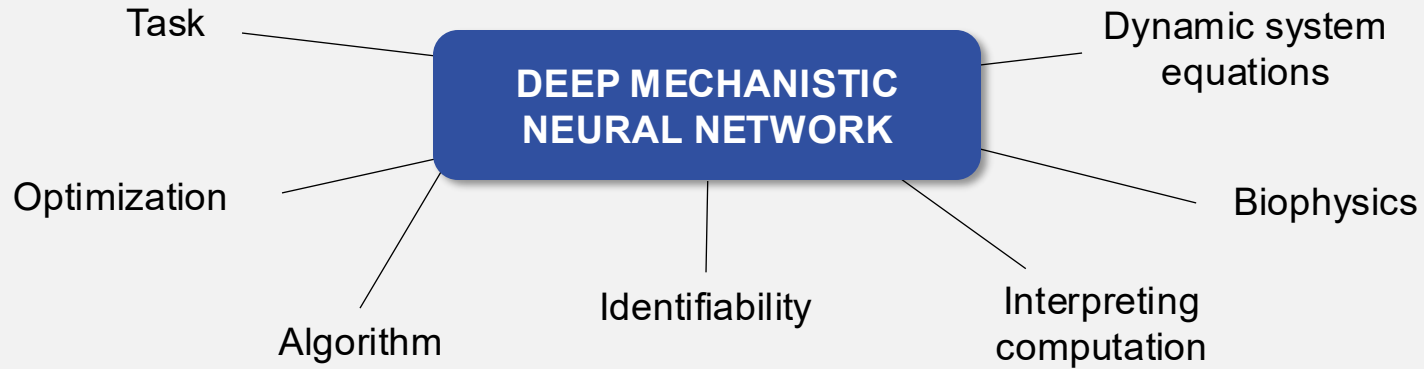


imprs-is



DEEP LEARNING

**COMPUTATIONAL
NEUROSCIENCE**



PROBABILITY

NEUROSCIENCE

Pretrained models and documented software



Flyvis Documentation

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Flyvis Documentation

A connectome-constrained deep mechanistic network (DMN) model of the fruit fly visual system in PyTorch.

- Explore connectome-constrained models of the fruit fly visual system.
- Generate and test hypotheses about neural computations.
- Try pretrained models on your data.
- Develop custom models using our framework.

Flyvis is our official implementation of [Lappalainen et al., "Connectome-constrained networks predict neural activity across the fly visual system." Nature \(2024\).](#)

Usage Guide

To get started we recommend going through our tutorials. These will guide you through the core concepts and provide practical examples:

Tutorials

1. [Explore the Connectome](#): Learn about the structure of the fly visual system connectome.
2. [Train the Network](#): Understand how to train the network on an optic flow task.
3. [Flash Responses](#): Explore how the model responses to flash stimuli.
4. [Moving Edge Responses](#): Analyze the model's responses to moving edge stimuli.
5. [Ensemble Clustering](#): Learn about clustering ensembles of models.
6. [Maximally Excitatory Stimuli](#): Discover how to find stimuli that maximally excite neurons.
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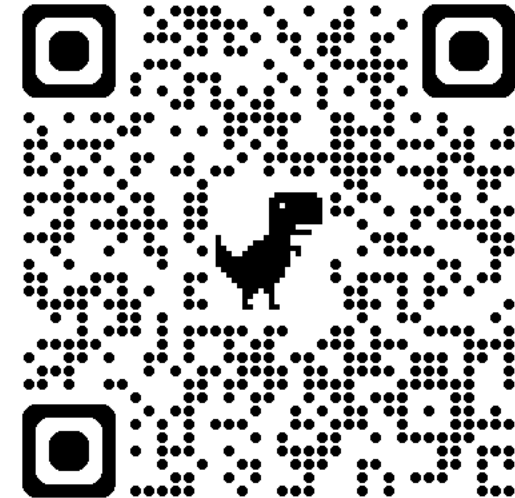
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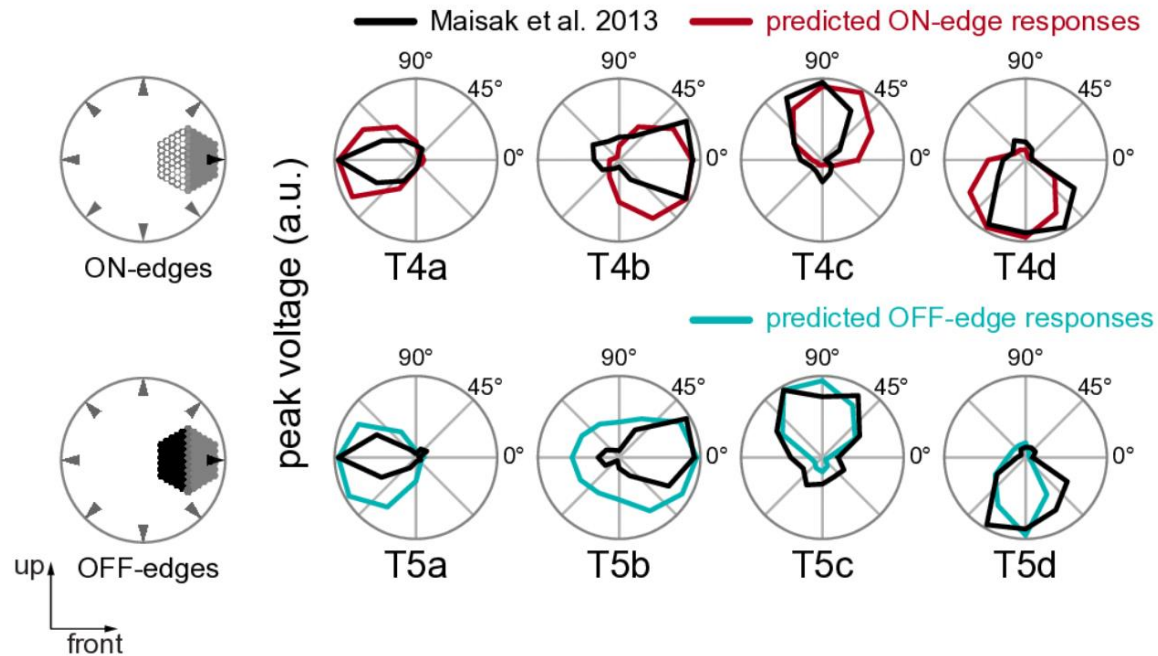
Project ideas

Summary

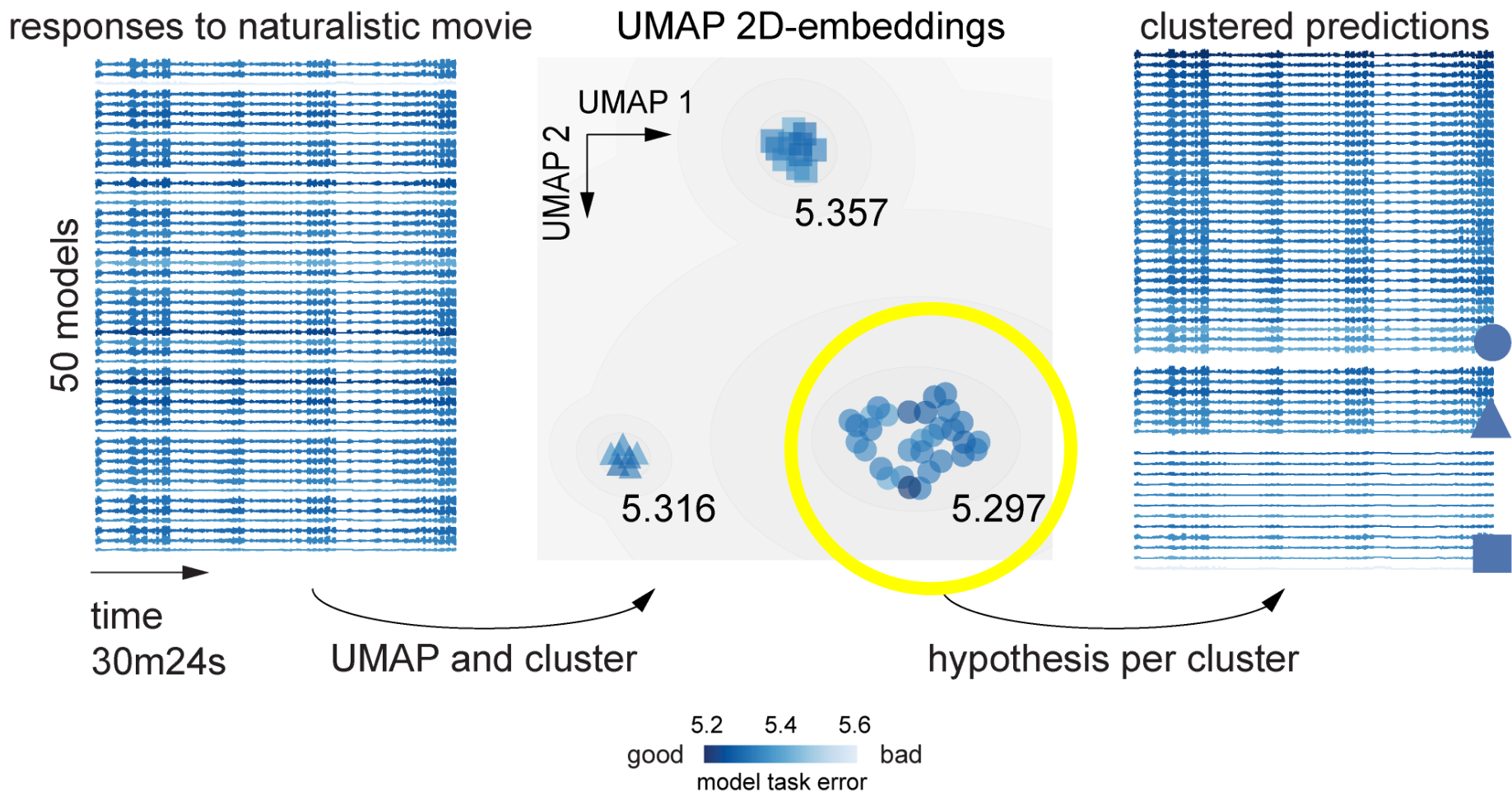
Core Results

Intro

Project Idea 1: Distilling ensemble connectome models into interpretable single networks via teacher-student training



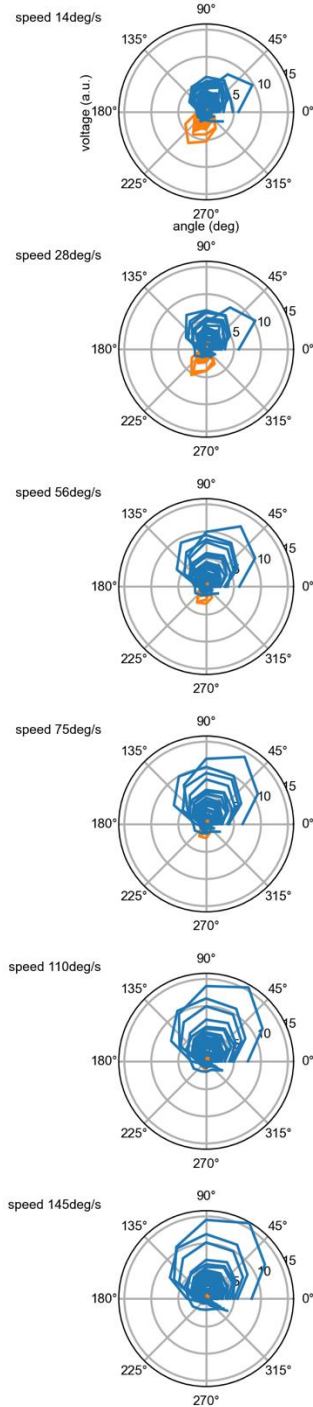
T4c



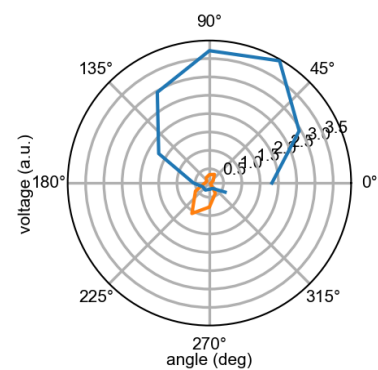


T4c

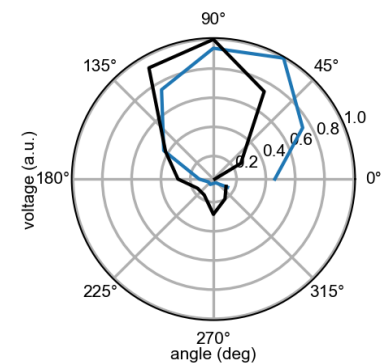
-> angular



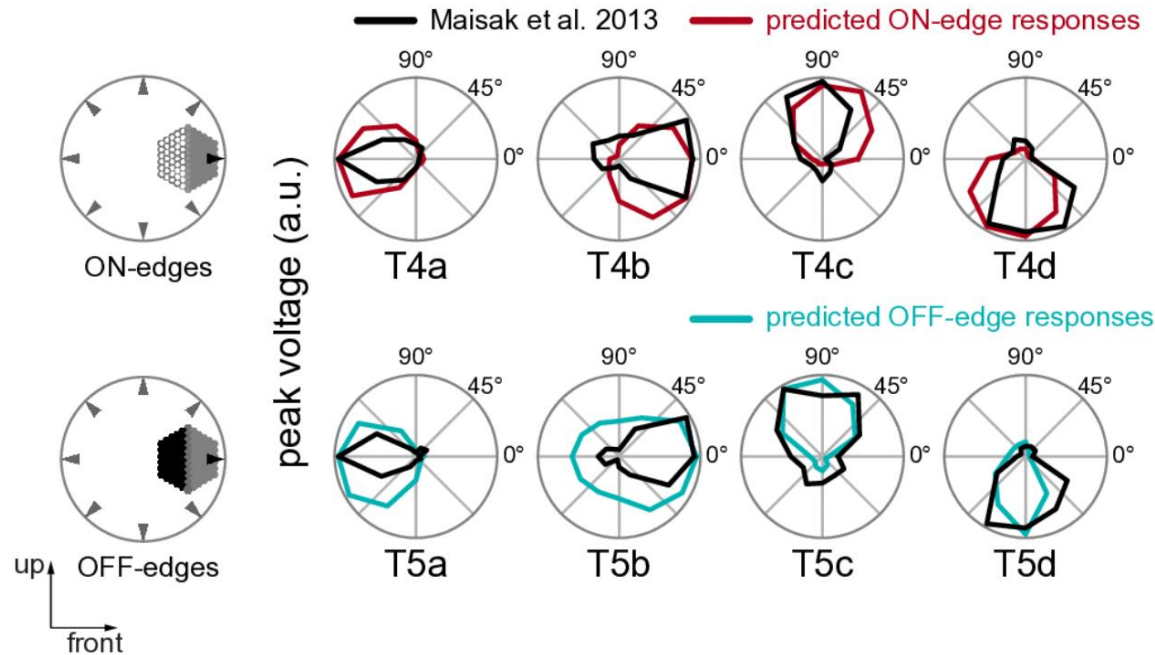
-> speed
and
model
average



-> max
abs scale



Project Idea 1: Distilling ensemble connectome models into interpretable single networks via teacher-student training



Ensemble Clustering

$$\{\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_K\} = \text{Cluster}(\{V_i^{(s)}(t)\}_{s=1}^M)$$

$$\mathcal{M}_{\text{best}} = \arg \min_k \frac{1}{|\mathcal{M}_k|} \sum_{s \in \mathcal{M}_k} \mathcal{L}_{\text{task}}^{(s)}$$

Teacher Construction

$$\mu_i^{\text{teacher}}(t) = \frac{1}{|\mathcal{M}_{\text{best}}|} \sum_{s \in \mathcal{M}_{\text{best}}} \frac{V_i^{(s)}(t)}{E[|V_i^{(s)}|]}$$

$$\hat{\mathbf{Y}}^{\text{teacher}} = \frac{1}{|\mathcal{M}_{\text{best}}|} \sum_{s \in \mathcal{M}_{\text{best}}} \hat{\mathbf{Y}}^{(s)}$$

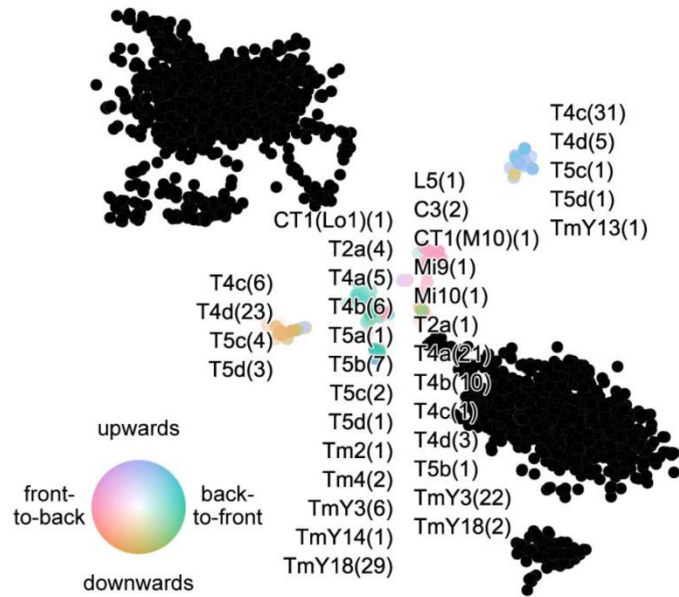
Student Training

$$\theta^{\text{student},*} = \arg \min_{\theta} \mathcal{L}_{\text{distill}} + \lambda \mathcal{L}_{\text{task}}$$

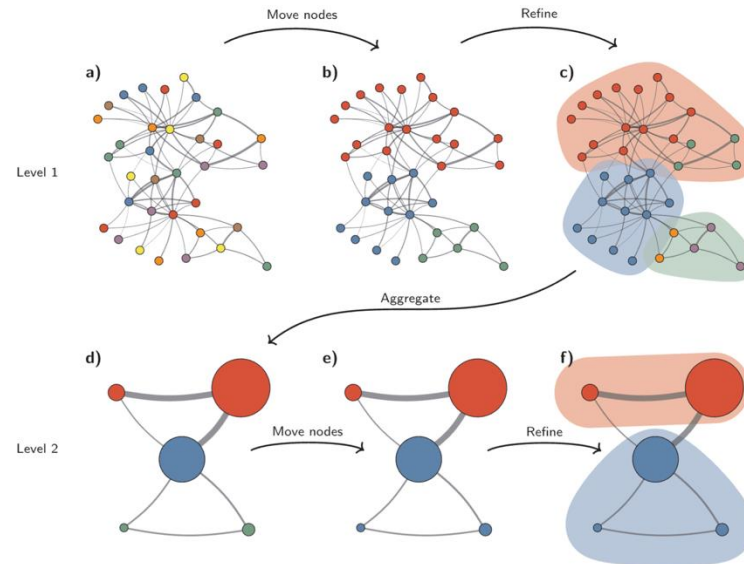
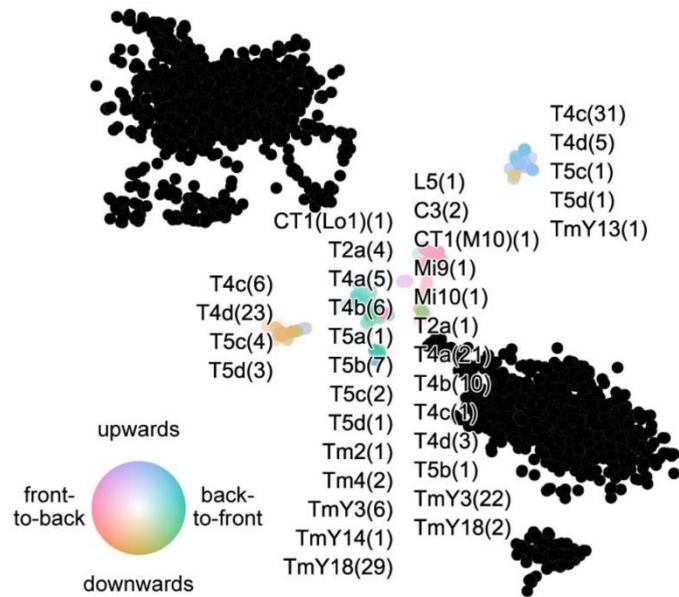
$$\mathcal{L}_{\text{distill}} = \|\mathbf{V}_i^{\text{student}} - \mu_i^{\text{teacher}}\|^2$$

$$\mathcal{L}_{\text{task}} = \|\hat{\mathbf{Y}}^{\text{student}} - \mathbf{Y}\|$$

Project Idea 2: Identify functional and mechanistically coupling cell-type communities

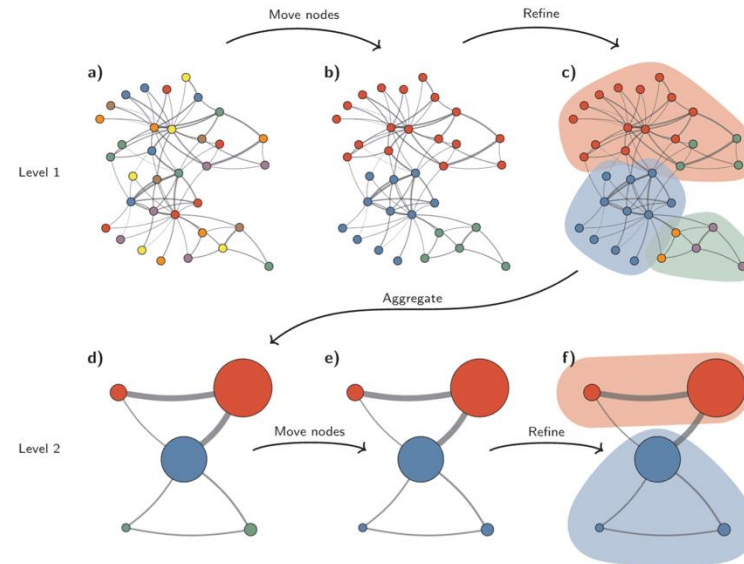
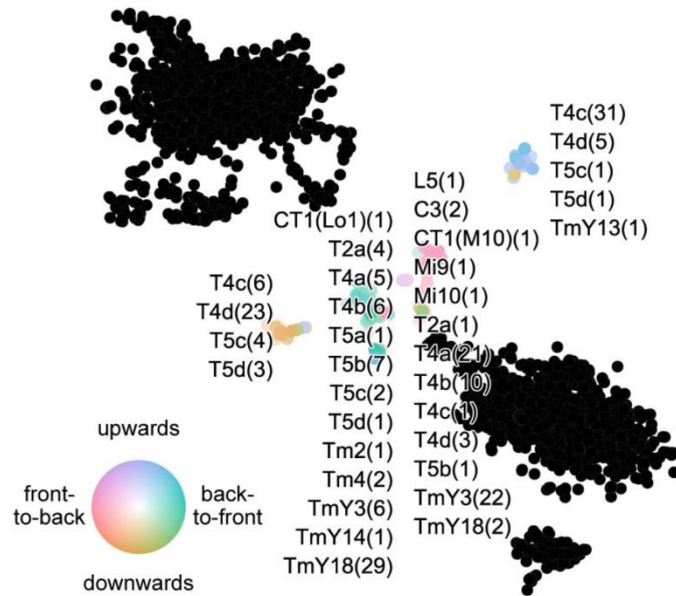


Project Idea 2: Identify functional and mechanistically coupling cell-type communities



Traag, V. A., Waltman, L. & van Eck, N. J. From Louvain to Leiden: guaranteeing well-connected communities. *Sci. Rep.* **9**, 5233 (2019).

Project Idea 2: Identify functional and mechanistically coupling cell-type communities



Community Detection Approaches

Functional: $\{\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_K\} = \text{Leiden}(\text{ResponseSimilarity}(\mathbf{R}^{(s)}))$

Connectivity: $\{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_L\} = \text{Leiden}(\mathbf{W}^{(s)})$

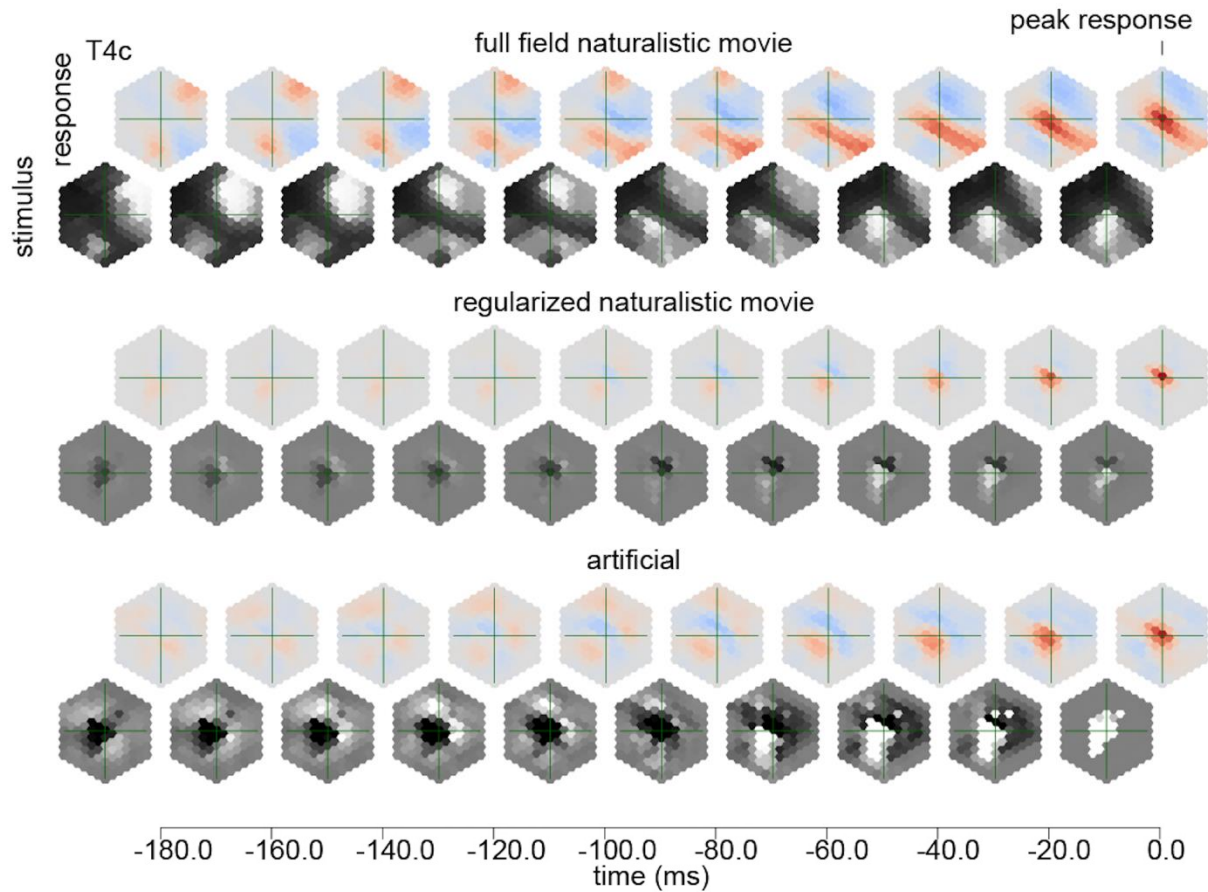
Cross-Space Validation

$$\text{Overlap}(\mathcal{F}_i, \mathcal{C}_j) = \frac{|\mathcal{F}_i \cap \mathcal{C}_j|}{|\mathcal{F}_i \cup \mathcal{C}_j|}$$

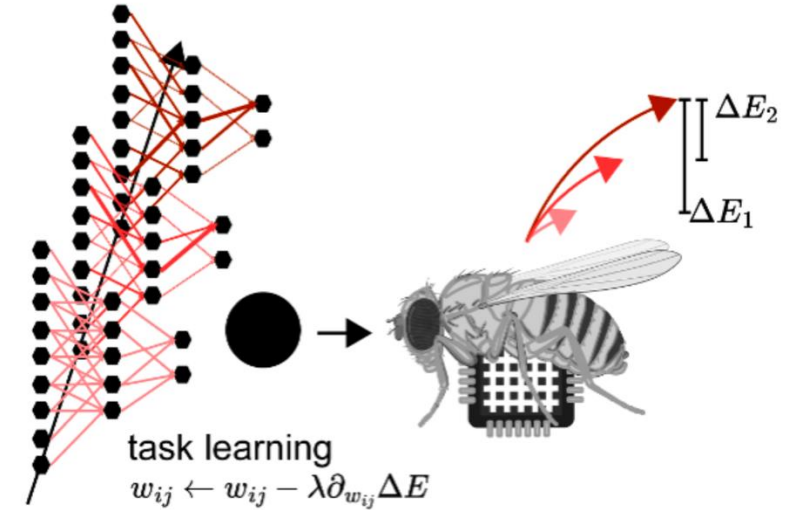
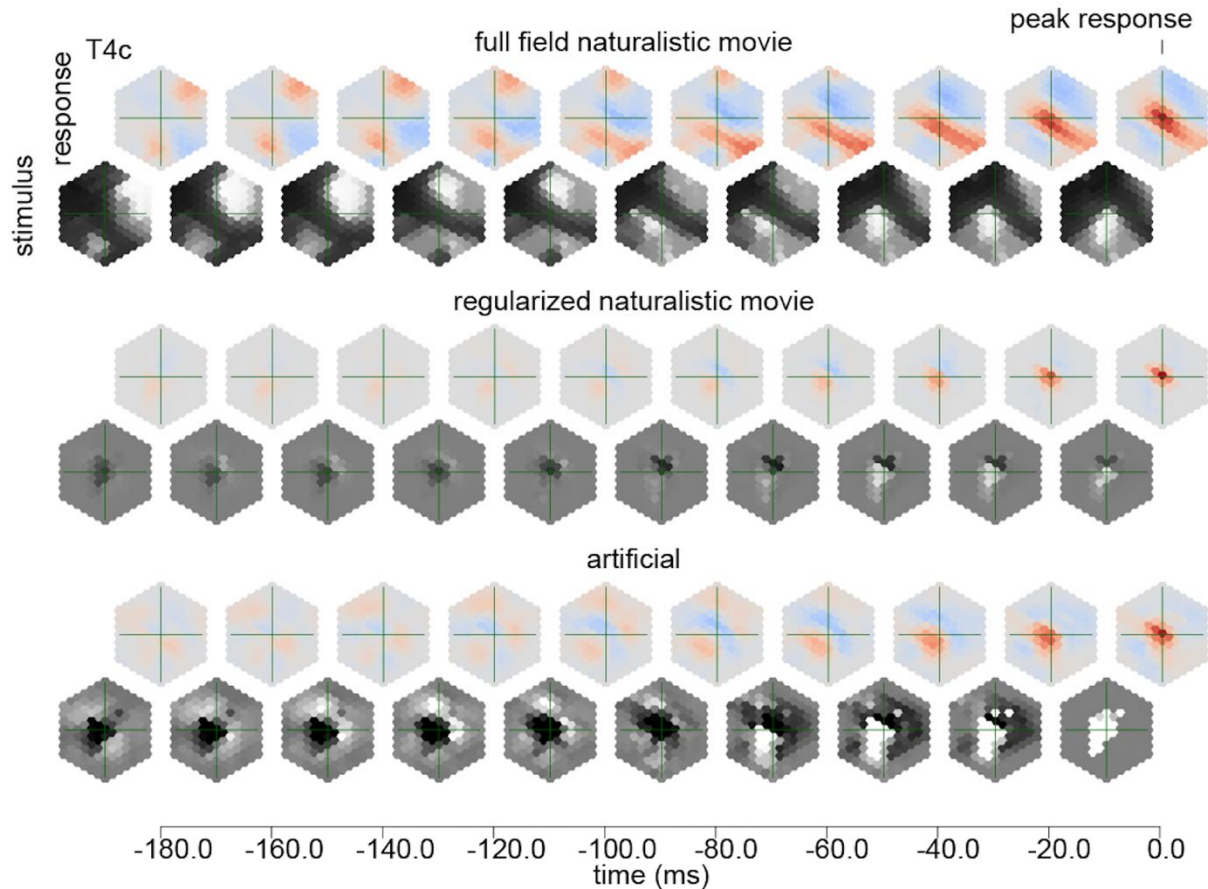
$$\text{Coherence}(\mathcal{C}_j) = \frac{1}{|\mathcal{C}_j|(|\mathcal{C}_j| - 1)} \sum_{i \neq k \in \mathcal{C}_j} \text{Corr}(\mathbf{R}_i^{(s)}, \mathbf{R}_k^{(s)})$$

Traag, V. A., Waltman, L. & van Eck, N. J. From Louvain to Leiden: guaranteeing well-connected communities. *Sci. Rep.* **9**, 5233 (2019).

Project Idea 3: Connect model to behavior and fool it with visual illusions



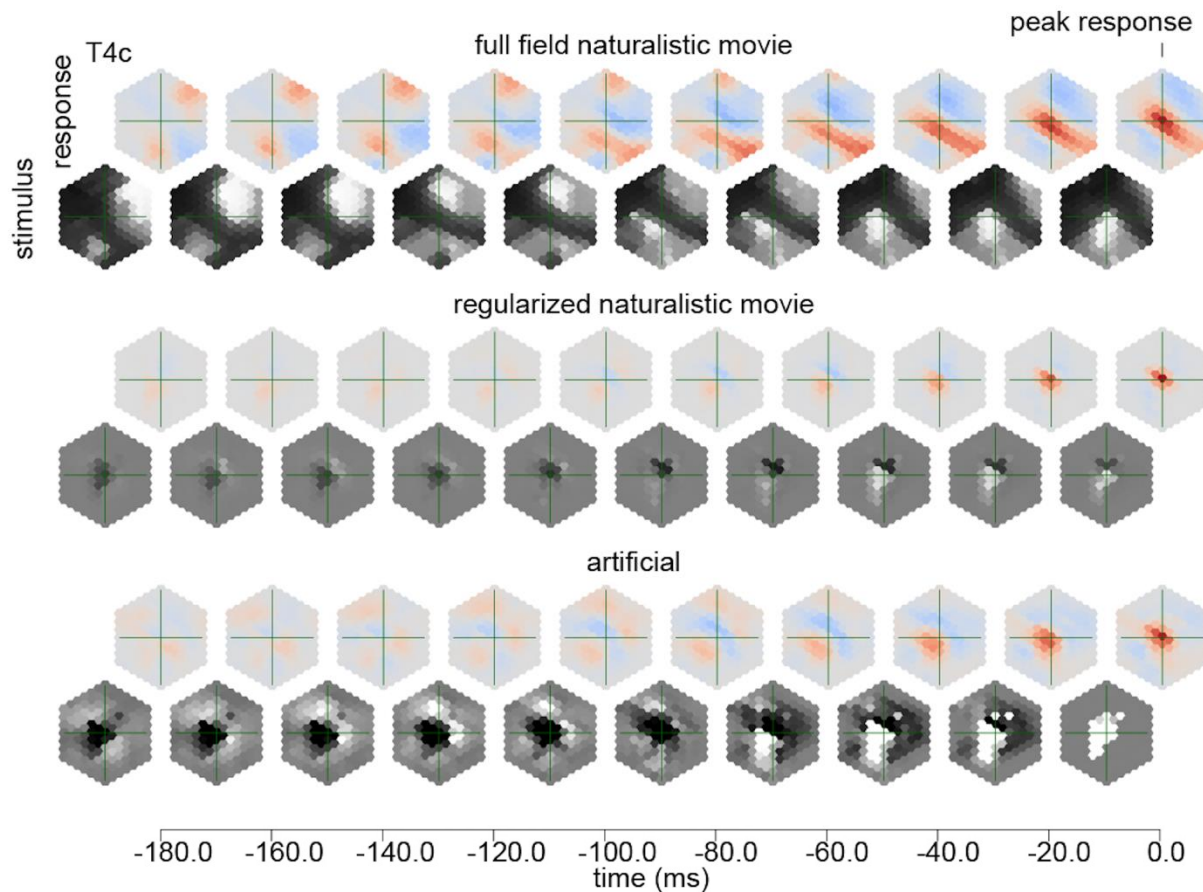
Project Idea 3: Connect model to behavior and fool it with visual illusions



Card, G. & Dickinson, M. H. Visually mediated motor planning in the escape response of *Drosophila*. *Curr. Biol.* **18**, 1300–1307 (2008).

Salem, W., Cellini, B., Frye, M. A. & Mongeau, J.-M. Fly eyes are not still: a motion illusion in *Drosophila* flight supports parallel visual processing. *J. Exp. Biol.* **223**, jeb212316 (2020).

Project Idea 3: Connect model to behavior and fool it with visual illusions



Behavioral Decoder Training

$$\mathbf{V}_{\text{looming}} = \text{Model}(\mathbf{X}_{\text{looming}})$$

$$\mathbf{V}_{\text{control}} = \text{Model}(\mathbf{X}_{\text{control}})$$

$$\hat{p}_{\text{looming}} = \text{Decoder}_{\text{behavior}}(\mathbf{V})$$

$$\phi^* = \arg \min_{\phi} \mathcal{L}_{\text{behavior}}$$

$$\mathcal{L}_{\text{behavior}} = - \sum_i [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)]$$

Illusion Generation

$$\mathbf{X}^* = \arg \min_{\mathbf{X}} \mathcal{L}_{\text{illusion}} + \lambda_1 \mathcal{L}_{\text{perceptual}} + \lambda_2 \mathcal{L}_{\text{smooth}}$$

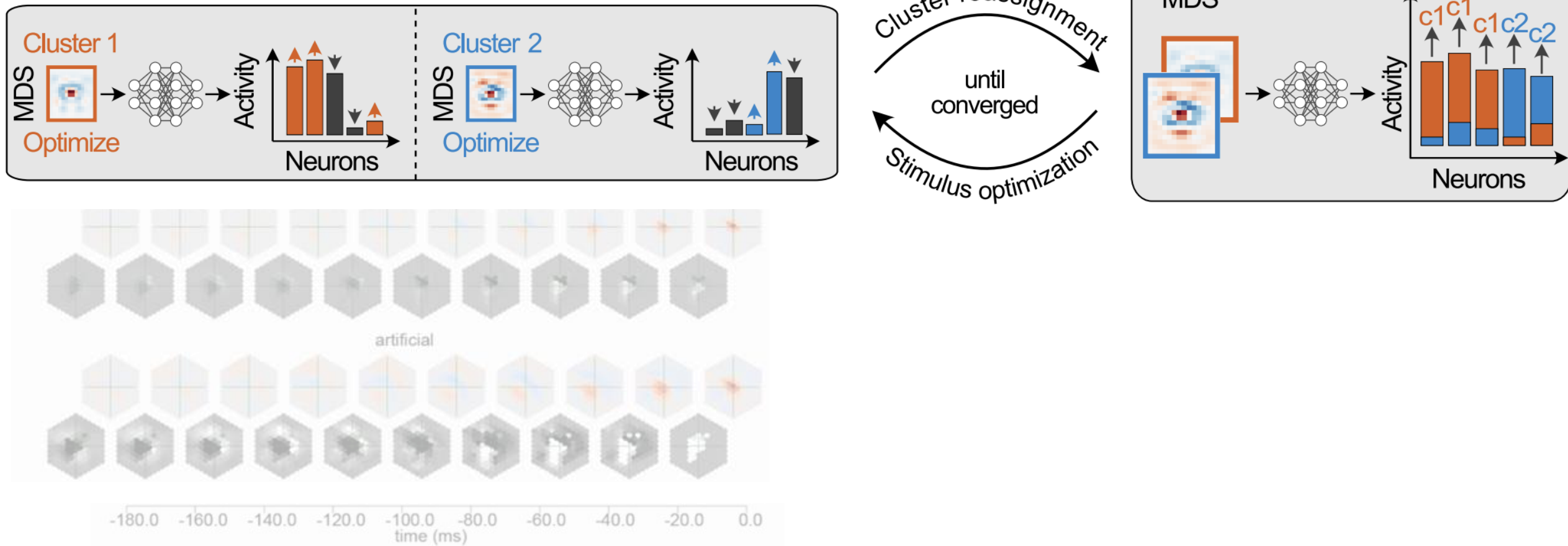
$$\mathcal{L}_{\text{illusion}} = \|\text{Decoder}_{\text{behavior}}(\text{Model}(\mathbf{X})) - y_{\text{target}}\|^2$$

$$\mathcal{L}_{\text{perceptual}} = \|\mathbf{X} - \mathbf{X}_{\text{base}}\|^2$$

$$\mathcal{L}_{\text{smooth}} = \|\nabla \mathbf{X}\|^2$$

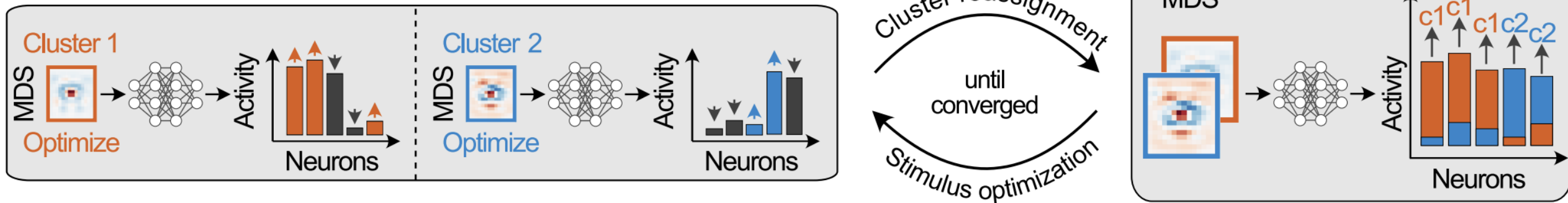
Project Idea 4: Create most discriminative stimuli for functional predictions across ensemble

B Maximally discriminative clustering algorithm



Project Idea 4: Create most discriminative stimuli for functional predictions across ensemble

B Maximally discriminative clustering algorithm



Algorithm 1 Most discriminative clustering algorithm

```

Randomly assign neurons into clusters
while cluster assignments change do
    M-step: Optimize MDS for all clusters with objective function Eq. (1)
    E-step: Re-assign each neuron to the cluster whose MDS drives it the most
end while
while cluster assignments change do
    for each cluster do
        Randomly assign this cluster's neurons into candidate sub-clusters
        Run EM clustering on sub-clusters until convergence
        Keep candidate sub-clusters if they improve mean objective  $\langle J_c \rangle_c$  and remove empty clusters
    end for
end while
return Cluster assignments, most discriminative stimuli
    
```

▷ Initial clustering

▷ Determine the optimal number of clusters



Project Idea 5: Deep mechanistic auxiliary targets

Goals:

1. **Biological plausibility:** Task-training produces arbitrary response scales and baselines inconsistent with for instance energy requirements or information rate transfers*
2. **Stimulus responsiveness:** Some neurons become silent to naturalistic stimuli
3. **Mechanistic control:** Direct shaping of internal dynamics and response properties
4. **Training efficiency:** (Pre)training with such unsupervised objectives may help task training

*Juusola, M. & de Polavieja, G. G. The rate of information transfer of naturalistic stimulation by graded potentials. *J. Gen. Physiol.* **122**, 191–206 (2003).

Project Idea 5: Deep mechanistic auxiliary targets

Goals:

- Biological plausibility**
response scales and biophysical instance energy requirements transfers
- Stimulus responsiveness**
silent to naturalistic stimuli
- Mechanistic control:** learn dynamics and response
- Training efficiency:** Biophysical training, not post-hoc analysis

Response Characterization

$$\begin{aligned}\mathbf{V}_{\text{grey}}^{(s)} &= \text{Model}^{(s)}(\mathbf{X}_{\text{grey}}) \\ \mathbf{V}_{\text{nat}}^{(s)} &= \text{Model}^{(s)}(\mathbf{X}_{\text{natural}}) \\ \mu_i^{(s)} &= E[\mathbf{V}_{i,\text{nat}}^{(s)}] \\ \sigma_i^{(s)} &= \text{Var}[\mathbf{V}_{i,\text{nat}}^{(s)}]\end{aligned}$$

Auxiliary Training

$$\begin{aligned}\theta^{\text{aux},*} &= \arg \min_{\theta} \mathcal{L}_{\text{aux}} \\ \tau_{t_i}^*, \alpha_{t_i t_j}^*, V_{t_i}^{\text{rest},*} &= \arg \min_{\tau_{t_i}, \alpha_{t_i t_j}, V_{t_i}^{\text{rest}}} \mathcal{L}_{\text{aux}}\end{aligned}$$

Auxiliary Target Design

$$\begin{aligned}\mathcal{L}_{\text{steady}} &= \sum_i \|\mathbf{V}_{i,\text{grey}}^{(s)} - \mu_{\text{target}}\|^2 \\ \mathcal{L}_{\text{scale}} &= \sum_i \left| \log \left(\frac{\sigma_i^{(s)}}{\sigma_{\text{target}}} \right) \right|^2 \\ \mathcal{L}_{\text{info}} &= \sum_i -I(\mathbf{V}_i^{(s)}; \mathbf{X}_{\text{natural}}) \\ \mathcal{L}_{\text{aux}} &= \lambda_1 \mathcal{L}_{\text{steady}} + \lambda_2 \mathcal{L}_{\text{scale}} + \lambda_3 \mathcal{L}_{\text{info}}\end{aligned}$$

Validation

$$\begin{aligned}\text{FRI}_i &= \frac{r_i^{\text{peak}}(I=1) - r_i^{\text{peak}}(I=0)}{r_i^{\text{peak}}(I=1) + r_i^{\text{peak}}(I=0)} \\ \text{DSI}_i &= \frac{|\sum_{\theta} r_i^{\text{peak}}(\theta) \exp(i\theta)|}{\sum_{\theta} r_i^{\text{peak}}(\theta)}\end{aligned}$$

